VECTOR



Release Notes
MICROSAR Classic Version 29.6

VAG SLP5

Document Version 29.06.07 | 2023-03-23

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Introduction

This document describes all changes and enhancements of features of the current and the previous two MICROSAR Classic releases.

Caution!

Although we carefully create this document, we have to highlight that this document has informal character only. Please consult the technical documentation of the delivery (e.g. Technical References and User Manuals) for a complete documentation on the delivered product.

The changes and enhancements of the last two development releases are listed in one chapter per development release. Former development releases are summed up. The changes and enhancements of the previous two MICROSAR Classic releases are also summed up in one chapter. The date stated in each heading corresponds to the release date of the respective MICROSAR Classic development release or MICROSAR Classic release.

Each chapter contains the following sections:

- **Implemented Features (incl. Migration Steps)**

  This section describes the newly implemented features grouped by functionality. Reading this section is recommended to find out which features have been added.

  

  If the feature requires the migration of existing projects, the migration steps are described in the feature description. Migration steps are highlighted with the symbol on the left.

  In addition, the migration steps are also provided in the sections Migration Step.

- **Migration Steps**

  This section describes all known changes required for migrating an existing project from one release to another. When starting with a new project, this chapter is not relevant. Reading this section is recommended while migrating an existing project to the new release.



Cross Reference

The latest version of this document is available at https://portal.vector.com/web/pes-dokumente/shared-folder?t=9c084b23-2f9a-4c60-8c5a-7aae3eccdadb.

Information on the roadmap of the Vector AUTOSAR solutions can be found here: https://portal.vector.com/web/pes-dokumente/shared-folder?t=5058cdcf-1dee-4c14-a1ab-d67817696aaa.

Release 29 (2022-09-23)

Implemented Features (incl. migration steps)

General

First and Fixed Version Removed From Issue Reports (ICM-1991)

The Issue Reports have been made more understandable by removing the information about ‘First affected version’ and ‘Fixed in versions’ reported for each issue. This information did no carry any relevant information as issues are only reported to affected deliveries a manual evaluation of this information was never required.

Note: This change will affect all issue reports provided from now on.

Introduction of Demo Components and Location of MakeSupport files (MSR4-10)

With R29, we introduce a new folder for Demo components in the MICROSAR BSW Package. It is located at external\DemoComponents.

- It contains all parts that Vector provides as an example and help for the integration. It is without any granted quality measures.
- Those components are for example: vBrs, vBaseEnv, vLinkGen, vItaHlp, Mcal Stub Modules, etc.

The MakeSupport files are placed directly at external\MakeSupport.

MemCpy with Endianness Conversion Functions Supported in VStdLib (MSR4-32048)

VStdLib now supports the following MemCpy functions which copy the source data with a converted endianess to the destination:

- MemCpySwapEndian (always converts src data to opposite endianess)
- MemCpyEndianPlatformToBig/MemCpyEndianBigToPlatform (no conversion if platform is big endian)
- MemCpyEndianPlatformToLittle/MemCpyEndianLittleToPlatform (no conversion if platform is little endian)

Furthermore, the following functions are supported:

- Large Macro for each API to support extended range for byte count.
- _s function wrapper for each API and Large Macro to support destination size check (returns errorCode, if destination size check fails otherwise E_OK)

VStdLib Supports MemCheck APIs (MSR4-33093)

The VStdLib provides new APIs to check that a memory area consists only of a pattern for a defined number of bytes. It returns E_OK to indicate that the content of the memory

General

area matches the pattern, else returns the errorCode passed as argument to indicate a mismatch.

The new APIs are VStdLib_MemCheck() and VStdLib_MemCheckLarge().

AUTOSAR Monitoring and Debugging

Support of Harware Trace Implementation via ARTI (MSR4-30237)

The new module ARTI (AUTOSAR Runtime Interface) according AUTOSAR 4.4 is available. ARTI is the successor of ORTI and uses the new introduces OS Hooks. ARTI allows a standardized trace implementation for major trace and debugging vendors for finding and solving performance issues and to perform timing measurements.

Charge

veScc Add-On BPT Available as Production QM According To ISO15118-20 (MSR4-1441)

The veScc Add-On BPT according to the ISO15118-20 is now available as Production QM.

veScc Add-On AC and DC Available as Production QM According To ISO15118-20 (MSR4-29611)

The veScc Add-Ons AC and DC according to the ISO15118-20 are now available as Production QM.

veSCC Provides Detailed Error Information as Return Type (MSR4-32897)

DetailedError already allows the application to get a detailed picture about what went wrong inside the CHARGE stack. However, some ReturnTypes are not used, instead it's only checked whether it's not equal to E_OK.

Now all ReturnTypes are utilized, so that they are reflected in the DetailedError description.

Application Interface To Report Auxiliary Power Supply Status in veCanCcGbt (MSR4-33472)

The module veCanCcGbt now provides the function (application callout) AppVCanCcGbt_IsAuxiliaryPowerSupplied that returns whether the auxiliary power is still supplied or not. The related actions based on "auxiliary power supply" defined in GB/T 27930-2015 are triggered accordingly.

Communication

CAN-FD J1939 support of extended TPs (Beta) (MSR4- 31196)

Communication

Based on BAM and CMDT a TP frame can now contain up to 64 bytes and direct frames are using IPduM containers. Note that this is functionality is a beta version.

Support of Multiple TCP Connections With Same Local IP Address and Port (MSR4-29098)

TcpIp now supports the parallel usage of multiple TCP sockets configured with the identical local endpoint (local IP address and local port). This allows a mix of (multiple) TCP clients and up to one TCP server with the same local endpoint. Multiple TCP servers with the identical local endpoint must not be configured.

Transferred to the SoAd module, this allows the configuration of multiple socket connections in a socket connection group for TCP client and server. Furthermore, different TCP client SoConGrps and a TCP server SoConGrp may share the same local address and port.

Handling of Cable Diagnostic According To AUTOSAR 20-11 (MSR4-29827)

The handling of cable diagnostic in EthSwt and EthTrcv has been completed and aligned to AUTOSAR 20-11.

Advanced Multi-Core Distribution for CAN (Beta) (MSR4-30277)

Multi-Partition now supports mapping of different CAN channels to different partitions as Beta to achieve load-balancing. Note that the following features are not yet supported:

- Mixed ASIL (all partitions require the same ASIL level),
- J1939 modules and Mirroring.

Safety Requirements (MSR4-303)

Time synchronization fulfills now safety requirements with ASIL D qualification.

Reporting of Security Events by TcpIp To IdsM (MSR4-30401)

TcpIp now reports the following security events to IdsM:

- TCPIP_SEV_ARP_IP_ADDR_CONFLICT: Is triggered in case of received local IP address in ARP reply for different MAC.
- TCPIP_SEV_DROPPED_INVALID_SOCKET_TCP_PACKET: Is triggered in case of dropped TCP packet because of invalid TCP-Socket.
- TCPIP_SEV_DROPPED_INVALID_SOCKET_UDP_PACKET: Is triggered in case of dropped UDP packet because of invalid UDP-Socket.
- TCPIP_SEV_DROPPED_INVALID_IPV4_ADDRESS_DATAGRAM: Is triggered in case of dropped datagram because of invalid IPV4 address for this controller.
- TCPIP_SEV_DROPPED_INVALID_IPV6_ADDRESS_DATAGRAM: Is triggered in case of dropped datagram because of invalid IPV6 address for this controller.

Communication

Synchronized PNC Shutdown Supported (MSR4-31267)

Synchronized PNC shutdown is now supported as specified in AUTOSAR R21-11 as draft. If this feature is activated, it enables to shutdown a PNC synchronously in the entire PNC topology. For this, ComM now provides the mechanism to request (as a top-level PNC coordinator) and forward (as an intermediate PNC coordinator) the synchronized shutdown. In addition, Nm and the bus-specific Nm modules now support the synchronized shutdown by indicating the request for a synchronized shutdown of a PNC to Nm or ComM, respectively.

LinTp Add-On AutoAddressAssignment Available as Production QM (MSR4-31699)

The LinTp Add-On AutoAddressAssignment is now available as Production QM.

The LinTp Add-On AutoAddressAssignment provides the autonomous processing of LIN Auto Addressing sequences and LIN PID assignment sequences parallel on several LIN channels. The LIN Auto Addressing methods “Bus Shunt Method 1” (method ID 0x02) and “Bus Shunt Method 2” (method ID 0xF1) are supported. It provides an interface to a SWC to dynamically configure these sequences during runtime and to control the sequence handling, either by an RTE port interface or by a plain C interface. The requested schedule table modes are notified to the BswM. It is implemented as a complex device driver acting as an upper layer transport protocol module of PduR to use LinTp for diagnostic communication.



Documentation in BSW Package
For more information, refer to the Technical Reference MICROSAR Classic vLinAA.

ECU Name Fields Can Be Changed Including AAC in J1939Nm (MSR4-31792)

The ECU name fields including the Arbitrary Address Capable (AAC) bit can now be changed via the API J1939Nm_SetName() or NAME management messages.

IPv6 Without Extension Headers (MSR4-32049)

TcpIp now supports (optionally) to discard received IPv6 packets which contain extensions headers which are not configured.

CAN-FD J1939 with TP (MSR4-32234)

J1939 using Can-FD is now supported including J1939Tp extensions, placing of J1939Nm NM and CA PDUs in containers. Also, J1939Rm ACKM PDUs can be placed additionally into FBFF containers.

CanTp Supports Strict and Non-Strict Length Check at Once (MSR4-32605)

Generally the ISO 15765-2 specifies that a strict length check should be applied when a

Communication
frame comes in. But some customers and OEMs deviate form the ISO and need therefore a non-strict length check support. Now the CanTp supports both options depending on the required need in the project.
Duplicate ACKs Supported in Tcp (MSR4-33085) TcpIp now supports duplicate ACKs as defined in section 2 DUPLICATE ACKNOWLEDGEMENT of RFC5681.
TLS Server Functionality Supported in TcpIp Add-On Tls (MSR4-33087) TcpIp Add-On Tls now supports the TLS Server functionality specified in RFC 8449 Record Size Limit Extension.
Support of 3rd Party CAN Driver Combination (Beta) (MSR4-33785) The combination of a 3rd Party CAN driver with AR version 4.4.0 and a CMSR CAN driver is supported for NXP LLCE CAN and CMSR FlexCan4.
Preparation for J1939-22 Support of the Extended TPs Based on BAM and CMDT (Beta) (PERMIL-3209, PERMIL-3207, MSR4-31196)    **This feature requires the migration of existing projects:** In a first step for supporting extended TPs according J1939-22, the following new source files have been created: > J1939Tp_FdBam.c and .h > J1939Tp_FdCmdt.c and .h > J1939Tp_FdDirect.c and .h > J1939Tp_Utils.c and .h Furthermore, the following changes were made to the description: > /J1939Tp/J1939TpGeneral/J1939TpSupportFdCmdt > /J1939Tp/J1939TpGeneral/J1939TpSupportFdBam > /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpRxCanFdChannel > /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpRxPg/J1939TpFdRxDirectNPdu > J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpFdRxCmNPdu > /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpFdRxDtNPdu > /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpFdTxFcNPdu > Added J1939TP_PROTOCOL_FD_CMDT and J1939TP_PROTOCOL_FD_BAM to the /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpRxProtocolType > /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxCanFdChannel

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- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxPg/J1939TpFdTxDirectNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxPg/J1939TpFdTxDirectBcNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpFdRxFcNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpFdTxCmNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpFdTxDtNPdu
- Added J1939TP_PROTOCOL_FD_CMDT and J1939TP_PROTOCOL_FD_BAM to the /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxProtocolType

Note that J1939Tp needs a PduR in minimum version 21.02.01 and and an IpduM in minimum version 14.02.01.

Diagnostics

Float Datatype Export To DEXT for DID Data Elements (DEM) (DCM-2550, MSR4-29071)



This feature requires the migration of existing projects:

Datatype and encoding are now evaluated during CDD-/ODX-Import for DEM data elements. That means that Boolean, Float, and Array Datatypes are imported as such and not as Unsigned or Byte-Array. This is a breaking change which might require modifications on the port interfaces.

Support of User-Defined Memory - Non Unique DTCs (DCM-5412, MSR4-1230)



This feature requires the migration of existing projects:

If a DTC with the same UDS DTC-Number resides in more than one event memory, this DTC was only imported once. With DaVinci Configurator Classic version 5.25 SP1, it is imported into each event memory as a separate DTC. Note that these DTC have different short-names although they use the same UDS DTC-Number.

Support of User-Defined Memory - EDR and SSR (DCM-5690, MSR4-1230)



This feature requires the migration of existing projects:

With DaVinci Configurator Classic version 5.25 SP1, each event memory may have its own extended data records (EDR) and snapshot records (SSR). Until now, if a user defined memory contained an EDR (or SSR) with the same record number as was used in the primary memory, the EDR of the primary memory was used also with DTCs residing in the user defined memory. Now, each event memory will use its own defined

Diagnostics

EDR and SSR. This might result into changes if the EDR (or SSR) is defined differently in primary and user defined memory.

Migrate Freeze Frame Parameters in BSWMD To ASR 19-11 (DEM-6824)



This feature requires the migration of existing projects:

When a MICROSAR R28 or older configuration is opened in the DaVinci Configurator Classic a migration of the configuration can be executed.

Updating the configuration in DaVinci Configurator Classic after this migration with an input file like CDD or ODX may result in duplicated /Dem/DemGeneral/DemFreezeFrameRecordClass containers. Only some of these containers are referenced by a /Dem/DemGeneral/DemFreezeFrameRecNumClass/DemFreezeFrameRecordClassRef. The unreferenced DemFreezeFrameRecordClass containers can be deleted.

Migration of Freeze Frame Parameters To ASR 19-11 (DEM-6824)

The following parameters have been moved from Dem/DemGeneral/DemFreezeFrameRecNumClass to a Dem/DemGeneral/DemFreezeFrameRecordClass container.

- DemFreezeFrameRecordTrigger
- DemFreezeFrameRecordNumber
- DemFreezeFrameRecordUpdate

The container has to be referenced from Dem/DemGeneral/DemFreezeFrameRecNumClass/DemFreezeFrameRecordClassRef.

The naming of enum literals of the following parameters has changed from 'DEM_RECORD_TRIGGER_X' to 'DEM_TRIGGER_ON_X':

- Dem/DemGeneral/DemFreezeFrameRecordClass/DemFreezeFrameRecordTrigger
- Dem/DemGeneral/DemExtendedDataRecordClass/DemExtendedDataRecordTrigger
- Dem/DemGeneral/DemGlobalFreezeFrame/DemGlobalFreezeFrameRecordTrigger



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Diagnostics

containers are referenced by a /Dem/DemGeneral/DemFreezeFrameRecNumClass/DemFreezeFrameRecordClassRef. The unreferenced DemFreezeFrameRecordClass containers can be deleted.

Diagnostic Stack Supports Float Data Type (MSR4-1201)

The diagnostic stack was extended to support natively the float 32bit data type. This change affected the DCM, DEM, DlagXf and the DEXT for exporting floats.

Typed C/S Interfaces for Dem Supported (MSR4-29071)

The module Dem now supports AUTOSAR datatypes on C/S interface according to RfC 72896 Typed C/S Interface. Furthermore, the Vector-specific extension of float types (FLOAT, FLOAT_N) for C/S interface are also supported.

Support Parallel Diagnostic Communication From Different Testers in Default Session (MSR4-30836)

The Dcm supports now parallel diagnostic communication from different testers in the default session. This functionality is requested with AUTOSAR ticket AR-94195 and is essential for customers using UDS based OBD.

Important Information
If parallel service processing is enabled, Dcm_GetActiveProtocol() shall no longer be called from CDDs or SWCs.

NvDataInterfaces for Dcm Read/Write DID Operations Provided (MSR4-30842)

The possible Dcm DID interfaces have been extended for a mechanism to create NvDataInterfaces by extending DcmDspDataUsePort with USE_DATA_NV_DATA_INTERFACE.

Transmit and Receive OEM Payload Types for Internal DoIP (MSR4-32053)

DoIP now supports to transmit and receive OEM-specific and not supported payload types. In addition, DoIP now allows to notify a user about a reception event. Together, both enable a user to send Ack/Nacks also for vehicle-internal communication.

Secure Coding in DCM (MSR4-32245)

The DCM module offers a secure coding feature natively. It allows the manufacturer to control the type of data that is allowed to be written into the ECU, by only accepting to write signed data.

Secure Coding in DCM (MSR4-32245)

The DCM module offers a secure coding feature natively. It allows the manufacturer to control the type of data that is allowed to be written into the ECU, by only accepting to

Diagnostics
write signed data. No external modules are needed.
Supporting User-Defined Fault Memories According To ISO 14229-1 - Use-Case Individual Storage Triggers (MSR4-32569) The Dem module now supports individual configuration of event memory storage trigger per event memory.
Support AUTOSAR Array Datatypes for Dcm S/R Interfaces (MSR4-32689) The Dcm module now also supports the following data types: Uint16_N, Uint32_N, Sint16_N, Sint32_N, Float, Float_N.
n/k+1 Socket Optional for DoIP (MSR4-33125) ISO-13400 specifies to support a n/k+1 socket to reject additional tester connection attempts, where n/k is the number of concurrent TCP/TLS data connections supported by the DoIP entity. The +1 socket has now been made optional to reduce the memory consumption of the entire project.
DoIP Routing Activation on TCP To Detect Unused TLS Connections (MSR4-33126) DoIP now performs an alive check on the TLS connection upon reception of the routing activation request on the TCP socket if the tester is already connected on the TLS connection. With this, TLS connections are closed if unused.
DEXT Export of Diagnostic Class as SDG (MSR4-33614) For the DEXT export, it can now be defined if the diagnostic class name shall be exported as a SDG. There are the following possibilities: > No SDG at all. > The qualifier of the diagnostic class. > A user defined string.
Parameter protocolKind Configurable (MSR4-33633) The parameter protocolKind of the diagnostic protocol is now configurable and can be set via the Interface-attribute DiagnosticProtocolKind in the CDD. If no value is provided the qualifier of the CANdelaStudio Interface will be used instead.
Dcm Supports User Specific Authentication Role Encoding (MSR4-34018) When Dcm is processing a service 0x29 0x03 request, the role information is directly read from the certificate via KeyM_CertElementGet(). The Dcm expects the authentication role information in the certificate to be encoded according to the AUTOSAR format. However, in some ECU projects it may be necessary to use certificates in which the format of the authentication roles is user-defined.

Diagnostics

The Dcm can be configured to no longer call KeyM_CertElementGet() to read the role information, but rather the callout function Dcm_GetAuthenticationRoles(). The purpose of this API is to allow the application to read the role information directly from the certificate by itself using the KeyM parameters certId and certElementId provided by this function. The role information shall then be transformed to the AUTOSAR format and passed back to Dcm.

Ethernet

AUTOSAR TLS Server Available as Production QM (MSR4-29244)

The AUTOSAR TLS Server functionality in the TcpIp module is now available as production QM.

Manual IP Address Assignment in IPv6 (MSR4-29927)

TcpIp now supports manual address assignment during runtime via the API TcpIp_RequestIpAddrAssignment() as an alternative to the address assignment that is automatically triggered after the Ethernet link is up.

Support of Server Name Indication in Tls Client (MSR4-30152)

The Tls Client of the TcpIp Add-On Tls now supports Server Name Indication according to RFC6066.

Callout Notification for Discarded or Replaced ARP Table Entries (MSR4-32646)

In case of a full ARP table it is possible to trigger a configurable callout to inform the upper layer about the limited size of the ARP table.

This callout is triggered if:

- a new entry arrives while the table is full
- an existing entry was updated

Hardware Security Module

Component-Based Folder Structure in veHsm Deliveries (MSR4-31324)

The folder structure of veHsm deliveries has been aligned with MICROSAR Classic deliveries and are now component-based.

Measurement and Calibration

Support of ARTI Hooks in OS (MSR4-30234)

AUTOSAR R4.4 standardizes the ARTI (AUTOSAR Run-Time Interface) as successor of the former ORTI (OSEK Run-Time Interface). With this feature, the ARTI Makros and the

Measurement and Calibration

ARTI ECU Module are supported by the OS. ORTI is still available and can be stored in the ARTI ARXML as well.

Support C-API Access To Measurement Points in Rtm (MSR4-31201)

Rtm now supports the following C-API calls:

- Rtm_PrepareMPSettings (selects the measurement points to be active when measurement is started)
- Rtm_StartMeasurement (triggers a measurement start in next Rtm_MainFunction)
- Rtm_StopMeasurement (triggers a measurement stop in next Rtm_MainFunction)
- Rtm_ClearMeasurementResults (triggers clear of measurement results in next Rtm_MainFunction)
- Rtm_GetMeasurementItem (now supports all measurement points)

These API calls can be used instead of Xcp-based steering and reporting.

Memory

User Manual Memory Architecture Available (-)

The User Manual Memory Architecture is now available in the .\Doc\UserManuals folder of your MICROSAR Classic BSW Package.

This User Manual provides general background information about the Vector memory solutions and guidance to select the best solution for a specific hardware platform.

Configuration Handling in Fee FlexNor (MSR4-29803)

Fee FlexNor now provides the following features beyond the AUTOSAR standard:

- Support of operation in bootloader mode where only an incomplete block

configuration is provided.

- Handling of added and removed blocks as well as with blocks with updates payload sizes.
- Support of NvM callback.

NvM ReadAll Split in Phases (MSR4-313)

NvM now provides the possibility to split the ReadAll process in two phases:

- NvM_PreReadAll(): The first phase reads all NvM blocks configured with enabled NvMSelectBlockForPreReadAll.
- NvM_ReadAll(): The second phase reads all other blocks configured with enabled NvMSelectBlockForPreReadAll.

Memory



Documentation in BSW Package
For more information, refer to the Technical Reference NvM.

veMem Mode (MSR4-32811)

The veMem solution provides an optional (configuration) synchronous, minimal functionality mode to decrease code size and simplify integration. Use case: synchronous read, write, and erase access to internal NV RAM via veMem driver.

Runtime Environment

Support the Direct Connection of SenderReceiver and Calibration Ports (MSR4-28948)

With this feature it is possible to connect directly Sender-Receiver R-Ports of an application SWC with Calibration P-Ports of a calibration SWC.The interfaces Rte_Read, Rte_DRead and Rte_IRead APIs will then return the calibration constant. Previously this resulted in a generator error. Now it is allowed.

Functionality of Rte Add-On Let Extended (MSR4-30309)

The Rte Add-On Let (Logical Execution Time) now supports multiple instantiations, PR-ports, and IsEvent ExecutionOrderConstraints.

Moreover, the memory consumption and runtime of the fill flush routines was reduced.

Assignment of Runnables To Code Sections (MSR4-31332)

Rte now supports to assign runnable entities in the application SWCs to specific code sections (like PIM and calibration data) by using the configured name of swAddrMethod as specified in SWS_Rte_07194.

RTE and SomeIpXf Support Initial Value for Optional Elements (MSR4-31508)

According to AUTOSAR 20-11 the handling of initialization for optional elements for the RTE and SomeIpXf was defined. This functionality is now supported. When the inital value of a record element is marked as not available, the RTE will mark the element as not available as well in the defined availabiltyBitfield array of the implementation data type. Previously all elements needed to be specified as part of the init value when application data types were configured. Now elements can also be marked as missing to skip their serialization with SomeIpXf in case TLV is configured.

Creation of Single Schedule Table Supported in Rte Add-On Let (MSR4-31889)

The Rte Add-On Let (Logical Execution Time) now supports the creation of a single schedule table to trigger all LET-relevant entities and driver tasks. This enables a higher precision since there is only a single schedule table per core for the full system that

Runtime Environment
triggers all LET-related actions. Up to now, alarms and schedule tables have been created per task and therefore have not been synchronized.
A2L Element-Specific Descriptions for Structure Elements Supported in Rte (MSR4-31895) Rte now supports reading the data structure element-specific A2L descriptions and export them into the A2L file for each element independently.
Sub-Element and Leaf Mappings for NvBlockDataMappings Supported in Rte (MSR4-33017) Rte now supports Sub-Element and Leaf Element Mappings for NvBlockDataMappings as described in AUTOSAR_TPS_SoftwareComponentTemplate TPS_SWCT_01659, 11.5.5.4 NvBlockDataMapping in R21-11. The feature requires DiagXf. It cannot be configured in Davinci Developer Classic and thus requires that the NvBlock SWC is stored in the ApplicationComponents folder of the configuration.
Data Type Std_TransformerError Supported in SWC Code (MSR4-33033) Rte now supports the data type Std_TransformerError as specified in AUTOSAR R20-11. Std_TransformerError is a struct that contains the error code and the transformer class to which the error belongs. In previous releases of AUTOSAR data type Rte_TransformerError was used.
Rte Supports IRQ Lock Optimization Also for ASIL Partitions (MSR4-33105) Due to defensive programming, the RTE generator automatically created additional interrupt locks in the past. These interrupt locks produced additional runtime overhead which can now be reduced to free up runtime resources in ASIL projects. The change is enabled when the parameter RteOptimizeLocksInASILPartitions is enabled in the configuration. RTE Analyzer automatically lists APIs that are called from safe components and that are not protected with interrupt locks so that the customer can confirm manually that the call context is correct.
Internal Behavior BSWMD Files Include Can-Enter-Exclusive-Area Relations (MSR4-33108) The internal behavior BSWMD files now include the Can-Enter-Exclusive-Area relations as required by SWS_Rte_07251.
Option To Set SWC Exclusive Area Method To CUSTOM or NONE in Rte (MSR4-340) Rte now also supports to set the implementation method of SWC Exclusive Areas to

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CUSTOM or NONE. With this, a customer-specific or none exclusive area implementation can be used.

McSupportData Support in RTE Generator (RTE-5606, MSR4-30181)

Starting with R28, RTE generator now also supports McSupportData generation.

McSupportData is the measurement and calibration description that is specified in AUTOSAR and can be used to create A2L files with the DVMcDataConverter that is provided as part of DaVinci Configurator Classic.

Note
Davinci Configurator Classic 5.25 SP1 and McDataConverter 2.04 are required for the usage of the new A2L workflow with McSupportData.

This feature requires the migration of existing projects:

McSupportData is only generated when so called flat instance descriptors for the measurement or calibration elements are configured.

The solving action of the RTE1239 validation message can create the flat instance descriptors, if needed.

In case McSupportData shall be used, the Rte.a2l, Rte_MemSeg.a2l and Rte_XcpEvents.a2l files no longer need to be included in the Master.a2l file. The RTE elements are automatically added to the McData.a2l that is produced by DVMcDataConverter.

However, it is not required to use McSupportData. Instead, also the Rte.a2l files that are generated directly by the RTE generator can still be included in Master.a2l.

New features will only be added to McSupportData in the future, however.

Beware that also the handling for DVMcDataConverter changed in R28. The generated McData.a2l file is no longer only a snippet that needs to be included in the Master.a2l file. Instead, it is a complete a2l file with all relevant information that can be passed directly to other tools (e.g. ASAP2Updater).

The path to the input Master.a2l needs to be given in the project settings editor in DaVinci Configurator Classic.

Safety

Dlt Available for FFI Up To ASIL D (MSR4-30522)

Dlt is now available for Freedom from Interference up to ASIL D.

StbM Safety Use-Case Extended (MSR4-31414)

Safety

The functional safe scope of StbM has been extended by the following functionalities:

- Offset and rate correction for timebases.
- Usage of pure local time timebases.
- pDelay measurement.
- Rate correction.

Furthermore, the supported ASIL level of the functional safe scope of StbM has been increased from ASIL B to up to ASIL D.

FlexRay Support for Time Synchronization With Integrity (MSR4-31416)

Fr and FrIf now support time synchronization with integrity according to TSR-112409.

vDoIPInt Available for FFI Up To ASIL D (MSR4-31620)

vDoIPInt is now available for Freedom from Interference up to ASIL D.

Security

Improved Validation of Certificate Configurations (MSR4-31215)

Earlier the configuration of certificates in the KeyM module was error-prone. This features tries to overcome this issue by simplifying the configuration by adding validations to evaluate the correctness of configured certificates. The validation is able to compare the configured certificate with either an input file or binary certificate.

Support of Cryptographic Satellites (MSR4-31251)

Cryptographic hardware accelerators may be used directly from within an application.

KeyM Supports Deletion of Certificates (MSR4-31662, MSR4-33018)

The KeyM was extended to support the deletion of already installed certificates. This function can be called via KeyM_CertDelete API.

Storage of Certificates of Different Curves in One Container in KeyM (MSR4-31664)

KeyM now allows the storage of X509v3 certificates in one container, independent of the curve that was used to create them. Up to now, separate containers had to be created for each curve and certificate and a RAM buffer was also required for each container. Now, KeyM treats a container as a storage unit for X509v3 certificates and when evaluating the content, it analyzes which curve was used to create it and then processes it accordingly. The certificate storage in shared certificate slots is only supported for certificates with storage type RAM.

IdsM Supports Customs Sinks for Qsev (MSR4-32757)

Security

The IdsM can now be configured to provide a callout that provides custom handling of qualified security events (Qsev) which cannot be handled by standard sinks.

Support of Attribute Certificates (AC) in KeyM (MSR4-32801)

KeyM now also supports configuration, parsing, and validation of Attribute Certificates (AC) as defined in RFC5755. So far, Public Key Certificates (PKC) has been supported only.

Attribute Certificates are certificates that have only a short validity period and do not contain a public key. They are used to transport authorization information such as rights and roles.

IdsM Allows Customized Sev Context Data (MSR4-33032)

The IdsM was extended to provide a callback that allows custom provision and structure of SEV conetxt data. They can be either removed, reordered or added.

Support of Further PSK Cipher Suites for TLS Server (MSR4-33221, MSR4-33789)

The AUTOSAR TLS Server supports now additional PSK Cipher Suites such as TLS_PSK_WITH_AES_128_GCM_SHA256 and TLS_PSK_WITH_NULL_SHA256. The complete list can be found in the Technical Reference of the TCPIP module.

HMAC Supported in veHsm Hardware Accelerators (MSR4-33440)

HMAC is now supported for all available Hash algorithms of the veHsm Hardware Accelerator drivers.

SHA2-224, SHA2-384, SHA2-512 Supported in veHsm Hardware Accelerators (MSR4-33441)

The following Secure Hash Algorithms are now supported by the veHsm Hardware Accelerator drivers:

- SHA2-224,
- SHA2-384,
- SHA2-512.

Reply To CERTREQ if No Matching CA Is Provided (MSR4-33923)

If no certificate can be found based on the received CERTREQ, vIKE ignores the CERTREQ. Instead it transmits all available certificates that fit into a reply message, thus allowing the sender to validate the recipient of the CERTREQ with an alternate certificate.

Software Download

Vector Decompression Updated to vStreamProc 3.xx (MSR4-31585)

Software Download

Vector Decompression has been updated to vStreamProc 3.xx.

System

TimeBaseState API in StbM (MSR4-32830)

StbM now provides the new API Std_ReturnType StbM_GetMainTime() that can be used to get the main time tuple (TGsync, TVsync), status, and rate deviation from the submitted time-base.

Migration Steps

Communication

Preparation for J1939-22 Support of the Extended TPs Based on BAM and CMDT (Beta) (PERMIL-3209, PERMIL-3207, MSR4-31196)

In a first step for supporting extended TPs according J1939-22, the following new source files have been created:

- J1939Tp_FdBam.c and .h
- J1939Tp_FdCmdt.c and .h
- J1939Tp_FdDirect.c and .h
- J1939Tp_Utils.c and .h

Furthermore, the following changes were made to the description:

- /J1939Tp/J1939TpGeneral/J1939TpSupportFdCmdt
- /J1939Tp/J1939TpGeneral/J1939TpSupportFdBam
- /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpRxCanFdChannel
- /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpRxPg/J1939TpFdRxDirectNPdu
- J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpFdRxCmNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpFdRxDtNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpFdTxFcNPdu
- Added J1939TP_PROTOCOL_FD_CMDT and J1939TP_PROTOCOL_FD_BAM to the /J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpRxProtocolType
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxCanFdChannel
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxPg/J1939TpFdTxDirectNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxPg/J1939TpFdTxDirectBcNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpFdRxFcNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpFdTxCmNPdu
- /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpFdTxDtNPdu
- Added J1939TP_PROTOCOL_FD_CMDT and J1939TP_PROTOCOL_FD_BAM to the /J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxProtocolType

Note that J1939Tp needs a PduR in minimum version 21.02.01 and and an IpduM in minimum version 14.02.01.

Feature description: **Preparation for J1939-22 Support of the Extended TPs Based on BAM and CMDT (Beta) (PERMIL-3209, PERMIL-3207, MSR4-31196)**

Diagnostics

Float Datatype Export To DEXT for DID Data Elements (DEM) (DCM-2550, MSR4-29071)

Datatype and encoding are now evaluated during CDD-/ODX-Import for DEM data elements. That means that Boolean, Float, and Array Datatypes are imported as such and not as Unsigned or Byte-Array. This is a breaking change which might require modifications on the port interfaces.

➡ Feature description: **Float Datatype Export To DEXT for DID Data Elements (DEM) (DCM-2550, MSR4-29071)**

Support of User-Defined Memory - Non Unique DTCs (DCM-5412, MSR4-1230)

If a DTC with the same UDS DTC-Number resides in more than one event memory, this DTC was only imported once. With DaVinci Configurator Classic version 5.25 SP1, it is imported into each event memory as a separate DTC. Note that these DTC have different short-names although they use the same UDS DTC-Number.

➡ Feature description: **Support of User-Defined Memory - Non Unique DTCs (DCM-5412, MSR4-1230)**

Support of User-Defined Memory - EDR and SSR (DCM-5690, MSR4-1230)

With DaVinci Configurator Classic version 5.25 SP1, each event memory may have its own extended data records (EDR) and snapshot records (SSR). Until now, if a user defined memory contained an EDR (or SSR) with the same record number as was used in the primary memory, the EDR of the primary memory was used also with DTCs residing in the user defined memory. Now, each event memory will use its own defined EDR and SSR. This might result into changes if the EDR (or SSR) is defined differently in primary and user defined memory.

➡ Feature description: **Support of User-Defined Memory - EDR and SSR (DCM-5690, MSR4-1230)**

Migrate Freeze Frame Parameters in BSWMD To ASR 19-11 (DEM-6824)

When a MICROSAR R28 or older configuration is opened in the DaVinci Configurator Classic a migration of the configuration can be executed.

Updating the configuration in DaVinci Configurator Classic after this migration with an input file like CDD or ODX may result in duplicated /Dem/DemGeneral/DemFreezeFrameRecordClass containers. Only some of these containers are referenced by a /Dem/DemGeneral/DemFreezeFrameRecNumClass/DemFreezeFrameRecordClassRef. The unreferenced DemFreezeFrameRecordClass containers can be deleted.

➡ Feature description: **Migrate Freeze Frame Parameters in BSWMD To ASR 19-11 (DEM-6824)**

Diagnostics

Migration of Freeze Frame Parameters To ASR 19-11 (DEM-6824)

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➡ Feature description: **Migration of Freeze Frame Parameters To ASR 19-11 (DEM-6824)**

Runtime Environment

McSupportData Support in RTE Generator (RTE-5606, MSR4-30181)

McSupportData is only generated when so called flat instance descriptors for the measurement or calibration elements are configured.

The solving action of the RTE1239 validation message can create the flat instance descriptors, if needed.

In case McSupportData shall be used, the Rte.a2l, Rte_MemSeg.a2l and Rte_XcpEvents.a2l files no longer need to be included in the Master.a2l file. The RTE elements are automatically added to the McData.a2l that is produced by DVMcDataConverter.

However, it is not required to use McSupportData. Instead, also the Rte.a2l files that are generated directly by the RTE generator can still be included in Master.a2l.

New features will only be added to McSupportData in the future, however.

Beware that also the handling for DVMcDataConverter changed in R28. The generated McData.a2l file is no longer only a snippet that needs to be included in the Master.a2l file. Instead, it is a complete a2l file with all relevant information that can be passed directly to other tools (e.g. ASAP2Updater).

The path to the input Master.a2l needs to be given in the project settings editor in DaVinci Configurator Classic.

➡ Feature description: **McSupportData Support in RTE Generator (RTE-5606, MSR4-30181)**

Release 28 (2022-04-08)

Implemented Features (incl. migration steps)

General

Support of AUTOSAR MemMap BSW Module-Based Include Structure for Several Modules (MSR4-31533)

For this release, several BSW components changed their MemMap include structure to <Mip>_MemMap.h, and publish their memory sections via BSW-IMPLEMENTATION (either statically in BSWMD or dynamically via DVCFG Service).

The following components made the transition in alphabetical order:

AvTp, BswM, ComM, CRC, Dcm, DoIP, E2E, EcuM, EthIf, Etm, Exi, FrArTp, FrIf, FrTp, IdsM, J1939Rm, LibCv, LinIf/LinTp, MemIf, NmIf, Sd, SomeIpTp, TcpIp, veBsdAd, veDam, veDem42, vDiagSecAcc, veDns, veDoIPInt, veHttp, veIKE, veJson, veRtm, veScc, veSecPrim, veSlac, veLinAA, XmlSecurity

Documentation in BSW Package
Refer to the User Manual Memory Mapping in the .\Doc\UserManuals folder of your MICROSAR Classic SIP for more information.

Support of AUTOSAR Schema 21-11 (MSR4-31845)

AUTOSAR R21-11 ECU Extracts and Diagnostic Extracts can now be imported.

AUTOSAR Monitoring and Debugging

Monitoring Series ECUs in veRtm Available Up To ASIL D (MSR4-30891)

veRtm now provides features to track the resource consumption of an ECU in series cars. The result values are stored in NvM to be able to check at all time if there where resource management problems on the ECU. The following features are supported and can be used within ASIL software up to ASIL D:

- CPU Load with values min/avg/max.
- Task Response Times as percentile (histogram).
- Max value of Task Stack Consumption.

These values are stored on NvM shutdown.

ROM Consumption Reduced in Dlt (MSR4-32881)

The memory layout has been optimized to save ROM consumption in Dlt.

Charge

veScc Uses KeyM for Certificate Handling & AUTOSAR TLS Client instead of vTLS Client (MSR4-29622, MSR4-29623)

The Certificate management and handling that was previously in the veScc module was transferred to the KeyM module as new central certificate handling component defined by AUTOSAR. Additionally, the new AUTOSAR TIs in the Tcplp module is used now instead of the Vector-specific veTIs Client module. The interfaces and calls from the veScc were updated to reflect those new changes to be more compliant to the AUTOSAR architecture.



This feature requires the migration of existing projects:

As a result, veScc will only perform the certificate handling specified by the ISO15118, all additional handling e.g. for OEM-specific update mechanisms must now be implemented by the customer application using the KeyM module directly

AUTOSAR TIs Supports Additional Functionality for Charging Use Cases (MSR4-30433)

The AUTOSAR TIs functionality in the Tcplp module was extended to support the additional requirements for Charging use cases. The new solution deprecates and replaces the old veTIs Client module provided in the MICROSAR Classic Charge cluster.

Support of PAG 2019-07 HVLM Specification in veScc (MSR4-30442)

The veScc module now supports the following specifications:

- 2019.07.30 SCC-HVLM Signalschnittstelle Porsche
- 2019.07.22 Funktionsspezifikation HVLM-SCC Porsche V09

veScc Service Handling and TLS Master Secret (MSR4-31505)

The veScc module has been extended by the following features:

Service Handling:

- Application can now selected all services using a generic approach, regardless if they are free or not.
  - Certificate services can be pre-configured if it is allowed as non-free service.
- Multiple waits have been implemented during service discovery process to allow asynchronous processing (on e.g. other ECUs).

TLS Master Secret:

- The TLS master secret can be transmitted over existing UDP socket (from SDP) for debugging purposes if enabled in the configuration. This UDP frame will be automatically collected by CANoe and Wireshark to decode the TLS session and

Charge

display all EXI message elements.

Support of HVLM Interface V12 in veScc (MSR4-31920)

The veScc module now supports the HVLM interface specification V12.

EXI Encoded Certificate Installation Response Message API in veScc (MSR4-32146)

A new API Scc_ProcessCertificateInstallationRes is now provided in veScc that can process an EXI encoded certificate installation response message. This interface may be used to write contract certificate chains from the customer application to veScc.

Inform Application about Remaining Validity in veScc (MSR4-32177)

veScc now additionally informs the application about the result of the remaining validity of the contract certificate check, i.e. the remaining days are provided to the application.

Communication

Com_IpduGroupControl Removed (COM-2458, MSR4-31460)

The API Com_IpduGroupControl and the corresponding ECUC parameter ComOptimizedIPduGroupHandling was removed from Com. Now only the API Com_IpduGroupStart is left supported as functional option.



This feature requires the migration of existing projects:

For the migration of existing projects, a migration script is provided.

ComDescriptionRoutingCodeGeneration Removed (COM-2523)

The API ComDescriptionRoutingCodeGeneration has been removed from the Com.



This feature requires the migration of existing projects:

For the migration of existing projects, a script is delivered.

Channel-specific LinIf Main Functions (LIN-508)

LinIf now provides a separate LinIf main function for each LinIf channel as specified by AUTOSAR 4.3. This allows the mapping of the main function of each LinIf channel to different OS taks, allowing a better distribution of the LinIf system load.



This feature requires the migration of existing projects:

The previous LinIf_MainFunction does not exist anymore, but is replaced by 1 to n LinIf_MainFunction_<channelname> main functions.

Communication
Each of these new LinIf main functions must be mapped to an OS task again.
Stack State Handling in TcpIp (MSR4-29021) TcpIp now supports stack state handling of the states ONHOLD, STARTUP, and SHUTDOWN.
Support of Multiple TCP Connections With Same Local IP Address and Port (MSR4-29098) TcpIp now supports the parallel usage of multiple TCP sockets configured with the identical local endpoint (local IP address and local port). This allows a mix of (multiple) TCP clients and up to one TCP server with the same local endpoint. Multiple TCP servers with the identical local endpoint must not be configured. Transferred to the SoAd module, this allows the configuration of multiple socket connections in a socket connection group for TCP client and server. Furthermore, different TCP client SoConGrps and a TCP server SoConGrp may share the same local address and port.
AUTOSAR TLS Client Available as Production QM (MSR4-29136) The TLS Client according to AUTOSAR as TcpIp Add-On Tls is now available as Production QM.
Support of trusted_ca_keys Extension According To RFC 6066 in Tls Client (MSR4-30206) The AUTOSAR-based Tls Client now supports sending the trusted_ca_keys extension using cert_sha1_hash as defined in RFC 6066. The cert_sha1_hash is provided by the KeyM.
Safety Requirements (MSR4-303) Time synchronization fulfills now safety requirements with ASIL D qualification.
Multi-Partition Support of Com (MSR4-30323) Com now supports multi-partition as defined by AUTOSAR 19-11. Com can now provide a local instance in every partition where a network is mapped to. The individual Com instances offer the Rx signals and consume the Tx signals of the corresponding network (on the same partition). Com instances can also route signals via partition boundaries (Signal Gateway).
Variant Handling for SNI Configuration Switch in FrNm (MSR4-30398) The SNI (SourceNodeIdentifier) configuration switch in FrNm now supports variant handling.
Reporting of Security Events by TcpIp to IdsM (MSR4-30401)

Communication

Tcplp now reports the following security events to IdsM:

- TCPIP_SEV_ARP_IP_ADDR_CONFLICT: Is triggered in case of received local IP address in ARP reply for different MAC.
- TCPIP_SEV_DROPPED_INVALID_SOCKET_TCP_PACKET: Is triggered in case of dropped TCP packet because of invalid TCP-Socket.
- TCPIP_SEV_DROPPED_INVALID_SOCKET_UDP_PACKET: Is triggered in case of dropped UDP packet because of invalid UDP-Socket.
- TCPIP_SEV_DROPPED_INVALID_IPV4_ADDRESS_DATAGRAM: Is triggered in case of dropped datagram because of invalid IPV4 address for this controller.
- TCPIP_SEV_DROPPED_INVALID_IPV6_ADDRESS_DATAGRAM: Is triggered in case of dropped datagram because of invalid IPV6 address for this controller.

J1939-22 Supported (MSR4-31195)

J1939-22 (J1939 using CanFD) is now supported excluding J1939Tp extensions and placing of J1939Nm NM and CA PDUs in containers. Also, J1939Rm ACKM PDUs can only be placed in FEFF containers (i.e. using 29bit extended CAN ID).

Streaming-Based Transmission Via TCP (MSR4-31329)

Tcplp now supports streaming-based transmission of data over TCP as specified in AUTOSAR. This enables RAM optimization for the TCP buffer and performance optimization for the transmission. This is especially useful for utilizing a ethernet communication bandwith with a high dataload or for gateway routing use cases.

Top Down Service Configuration for ComM Users Supported (MSR4-31356)

ComM now supports Top Down Service Configuration for ComMgrUserNeeds as specified by AUTOSAR.

Fan-Out Use Case of SomeIpTp with IPsec Extended in SoAd (MSR4-31467)

The SoAd now provides an option to return E_OK on a fan-out if at least one of the socket connections could transmit the requested data. Up to now, the SoAd returned E_NOT_OK at SoAd_IfTransmit() if the transmission failed on at least one of the related socket connections. On fan-out of SomeIpTp messages, this prevented a transmission on the socket connection on which IPsec works properly.

I-PDU Callouts in Com (MSR4-31471)

Com now only supports I-PDU Callouts with PduInfoPtr (PduInfoType) according to AUTOSAR version 4.1.2 and later. The I-PDU Callout functions with a data pointer as a parameter are therefore no longer supported.

Reconnection of an Ethernet Communication Channel in EthIf (MSR4-31509)

EthIf now provides the possibility to silently re-connect to an Ethernet switched network

Communication

to avoid a wake-up across the network after an unexpected Ecu reset.

Up to now, an unexpected Ecu reset within an Ethernet switched network led to a situation that the affected Ecu lost its communication state which was established before the unexpected reset. Since a wake-up line is used for a switched Ethernet network, an ECU may not detect that communication is still needed.

Soft Activation Line State Change in DoIP (MSR4-31666)

DoIP now provides the mechanism to switch the Activation Line State to inactive within a configurable timeout. DoIP closes all socket connections used for diagnostics (i.e. managed by DoIP) until the timeout exceeded. This feature can be used instead of the already provided DoIP_Shutdown API which closes all socket connections independent of the use case.

LinTp Add-On AutoAddressAssignment Available as Beta (MSR4-31685)

The LinTp Add-On AutoAddressAssignment is now available as Beta.

The LinTp Add-On AutoAddressAssignment provides the autonomous processing of LIN Auto Addressing sequences and LIN PID assignment sequences parallel on several LIN channels. It provides an interface to a SWC to dynamically configure these sequences during runtime and to control the sequence handling. The requested schedule table modes are notified to the BswM. It is implemented as a complex device driver acting as an upper layer transport protocol module of PduR to use LinTp for diagnostic communication.



Documentation in BSW Package
For more information, refer to the Technical Reference MICROSAR Classic vLinAA.

LinTp Add-On AutoAddressAssignment Available as Production QM (MSR4-31699)

The LinTp Add-On AutoAddressAssignment is now available as Production QM.

The LinTp Add-On AutoAddressAssignment provides the autonomous processing of LIN Auto Addressing sequences and LIN PID assignment sequences parallel on several LIN channels. The LIN Auto Addressing methods “Bus Shunt Method 1” (method ID 0x02) and “Bus Shunt Method 2” (method ID 0xF1) are supported. It provides an interface to a SWC to dynamically configure these sequences during runtime and to control the sequence handling, either by an RTE port interface or by a plain C interface. The requested schedule table modes are notified to the BswM. It is implemented as a complex device driver acting as an upper layer transport protocol module of PduR to use LinTp for diagnostic communication.

Communication

Documentation in BSW Package
For more information, refer to the Technical Reference MICROSAR Classic vLinAA.

Rx-Path Supported for QSEv IDS-PDUs in PduR to Master-IdsM (MSR4-31713)

PduR now supports Rx-paths to forward QSEv IDS-PDUs from a Satellite-IdsM to a Master-IdsM.

UdpNmWaitBusSleepTime Extended (MSR4-31941)

The UdpNmWaitBusSleepTime has been extended beyond the AUTOSAR standard by the possibility to change the UserDefined parameter up to 200s.

Specific MSL for DoIP Sockets for TCP (MSR4-32036)

In SoAd, an MSL (Maximum Segment Lifetime) can now be configured for a specific DoIP TCP socket. With this, the time in the TIME-WAIT state could be reduced and a fast connection reestablishment is enabled.

Split of TcpIp Sections (MSR4-32038)

TcpIp now allows splitting of bigger data arrays to dedicated memory sections.

IPv6 Without Extension Headers (MSR4-32049)

TcpIp now supports (optionally) to discard received IPv6 packets which contain extensions headers which are not configured.

Option To Disable IPv4 UDP Checksum Calculation in TcpIp (MSR4-32103, TCPIP-4355)

TcpIp now supports disabling of IPv4 UDP checksum calculation in software at runtime. For this, the TcpIp_ChangeParameter() API has been extended by the option TCPIP_PARAMID_UDP_CHECKSUM as specified by AUTOSAR 4.3.0.

In case the UDP checksum calculation is disabled, the checksum of a transmitted frame is set to zero and received frames with checksum zero are not dropped but forwarded to the upper layer.

This feature requires the migration of existing projects:

With the introduction of this AUTOSAR feature, the handling of received/transmitted IPv4 UDP frames with checksum == 0 is changed: Received IPv4 UDP frames with checksum == 0 will be discarded by the TcpIp implementation and not forwarded to the upper layer.

To allow such frames, the functionality must be unlocked per socket:

Communication
> For SoAd as upper layer by disabling parameter: /SoAd/SoAdConfig/SoAdSocketConnectionGroup/SoAdSocketProtocol/SoAdSocketUdp/SoAdSocketUdpChecksumEnabled > For customer applications as upper layer by usage of TcpIp_ChangeParameter() API. If the functionality is unlocked, the TcpIp module will also use checksum == 0 for transmission of IPv4 UDP packets on the corresponding socket(s). See AUTOSAR requirements SWS_TCPIP_00279 and SWS_TCPIP_00280 and technical documentation for further details.
Setting the Frame Priority on Sockets for QNX in vBsdAd (MSR4-32221) vBsdAd now also supports setting the frame priority on sockets for QNX platforms.
Configuration of TcpIp RST Messages (MSR4-32382) TcpIp answers incoming Tcp messages for closed ports with an RST message as specified in RFC 793. Through this, RST message remote devices can scan for closed ports. It is now also possible to configure that no RST message is sent in this scenario. For this, DaVinci Configurator Classic provides a flag to enable/disable this functionality.
CanTp Supports Strict and Non-Strict Length Check at Once (MSR4-32605) Generally the ISO 15765-2 specifies that a strict length check should be applied when a frame comes in. But some customers and OEMs deviate form the ISO and need therefore a non-strict length check support. Now the CanTp supports both options depending on the required need in the project.
New Parameter vIKEConnectionDisablingTimeout in IKEv2 (MSR4-32699) IKEv2 of the TcpIp Add-On IPsec now provides a new parameter vIKEConnectionDisablingTimeout in the container /MICROSAR/vIKE/vIKEConnections with the following values: Min Value: 0s Max Value: 86400s Default Value: 4s Multiplicity 0..1 (optional) If this timeout is configured, it acts as an upper bound on the time vIKE will wait for the peer to respond to its delete requests. The timer starts as soon as VIKE_NO_ COMMUNICATION is requested. When it elapses, the connection is immediately torn

Communication

down and transitions into VIKE_NO_COMMUNICATION without further waiting for the peer.

Duplicate ACKs Supported in Tcp (MSR4-33085)

TcpIp now supports duplicate ACKs as defined in section 2 DUPLICATE ACKNOWLEDGEMENT of RFC5681.

TLS Server Functionality Supported in TcpIp Add-On Tls (MSR4-33087)

TcpIp Add-On Tls now supports the TLS Server functionality specified in RFC 8449 Record Size Limit Extension.

MemMap Usage in E2ELib (MWDG-5808, MSR4-31533)



This feature requires the migration of existing projects:

Background:

The E2ELib now uses the dynamically generated E2E_MemMap.h for the memory mapping, to reduce the compile time. E2E Library is non-generated, deterministic software code, where all inputs and settings are passed by function parameters. So in previous versions it was not possible to add it to the configuration. But now it provides a dummy BSWMD, which must be added to the configuration, that the MemMap generator recognizes it and generates the E2E_MemMap.h.

Migration:

To generate the E2E_MemMap.h the E2ELib must be added to the configuration manually.

Compatibility:

To be compatible with previous releases (before R27) the E2ELib still provides the E2E_MemMap.inc. In this case the memory sections are mapped via the MemMap.h which is included by a static E2E_MemMap.h. No migration is necessary.

Diagnostics

First Part of Supporting User-Defined Fault Memories According To ISO 14229-1 - Use-Case SEM (DEM-6320, MSR4-1230)

The DEM module now provides N numbers of user-defined Fault Memories according to AUTOSAR 4.4. and ISO 14229-1. In this first part all user-defined memories have the same properties and cannot be individually configured or their respective storages individually triggered. This will be done in the upcoming second part. Only property that can be set memory-specific is DemTypeOfFreezeFrameRecordNumeration. The DCM was extend in their service routines to allow access and authentication checks of the

Diagnostics

individual fault memories.

Current implementation supports the functional use-case of the of the Security Event Memory (SEM) concept as specified in AUTOSAR.



This feature requires the migration of existing projects:

It will no longer be possible to map internal events (events with no DTC reference) to primary/user-defined memories anymore. Hence, it will not be supported to clear internal events from primary and user-defined memories separately anymore.

For the purpose of clearing the events, all internal events will be virtually mapped to the event destination referenced by DemGeneral/DemClearEventsWithoutDTCEventMemoryRef. They can be cleared only by clearing the event destination referenced by DemClearEventsWithoutDTCEventMemoryRef.

While upgrading a configuration with internal events to this software revision, the Dem will add a default value for DemGeneral/DemClearEventsWithoutDTCEventMemoryRef. This value must be verified for its compatibility with your requirements.

MemMap Usage in vDam (HSM-2208, MSR4-31533)



This feature requires the migration of existing projects:

From now on, the MemMap sections can be generated into a file called vDam_MemMap.h. This requires including the Common_MemMap generator into the project and using the file vDam_MemMap.arxml to extract the used memory sections.

In order to keep compatibility to older projects, a file vDam_MemMap_Compatibility.h is provided in the implementation folder. During the SWCP, this file is renamed to vDam_MemMap.h and it relies on the file MemMap.h which was used until now.

Typed C/S Interfaces for Dem Supported (MSR4-29071)

The module Dem now supports AUTOSAR datatypes on C/S interface according to RfC 72896 Typed C/S Interface. Furthermore, the Vector-specific extension of float types (FLOAT, FLOAT_N) for C/S interface are also supported.

Support of Positive Response After Reset for Service EcuReset (0x11) in Dcm (MSR4-30049)

Dcm now supports to send a positive response after reset for service EcuReset (0x11) as defined by AUTOSAR. The response behavior can be configured in the container DcmDspEcuReset.

Receive Data in StartOfReception for DoIP-Dcm Use Case (MSR4-30634)

Diagnostics

Received data in the PduR callout StartOfReception is now provided to Dcm which now identifies the functional tester present within the scope of StartOfReception as specified in AUTOSAR.

NvDataInterfaces for Dcm Read/Write DID Operations Provided (MSR4-30842)

The possible Dcm DID interfaces have been extended for a mechanism to create NvDataInterfaces by extending DcmDspDataUsePort with USE_DATA_NV_DATA_INTERFACE.

vDiagSecAcc module specific MemMap support (MSR4-31533, SEC-3631)



This feature requires the migration of existing projects:

From now on, the MemMap sections can be generated into a file called vDiagSecAcc_MemMap.h. This requires including the Common_MemMap generator into the project and using the file vDiagSecAcc_MemMap.arxml to extract the used memory sections.

In order to keep compatibility to older projects, a file vDiagSecAcc_MemMap_Compatibility.h is provided in the implementation folder. During the SWCP, this file is renamed to vDiagSecAcc_MemMap.h and it relies on the file MemMap.h which was used until now.

UDS Service Authentication Role Modelling According to AUTOSAR (MSR4-31556)

The configuration for assignment of role to diagnostic services in Dcm is now explicitly modelled as ECUC container according to the updated AUTOSAR specification. With this, the configuration is now consistent with all other Dcm ECUC modelling. Up to now, the roles were decimal values, where each bit represents a role.

New Attribute Supported in DDM (MSR4-31875)

DDM now supports a Diagnostic Instance attribute for Routines, qualifier ASR3_Legacy_Combine_Routine, type unsigned. If value != 0, DDM creates one combined data element for the routine parameters, instead of individual elements. The data type of the combined data element is a byte array.

Custom Service Processors for UDS Service 0x29 in Dcm (MSR4-31942)

Dcm now allows to implement a custom service processor for service 0x29 (Authentication) or its subfunctions.

Gracefully Closing Tcp Connection During DoIP Negative Acknowledgment (MSR4-31991)

DoIP now closes a TCP connection gracefully (using FIN) in some cases instead of using RST. With this, testers can receive DoIP NACK messages which are sent shortly before the TCP connection is closed.

Diagnostics

Transmit and Receive OEM Payload Types for Internal DoIP (MSR4-32053)

DoIP now supports to transmit and receive OEM-specific and not supported payload types. In addition, DoIP now allows to notify a user about a reception event. Together, both enable a user to send Ack/Nacks also for vehicle-internal communication.

Data Element Size of Dem Extended (MSR4-32064)

Dem now supports data element sizes of up to 4096 Bytes.

Support of Different Payload Types for UUDT (MSR4-32165)

The module veDoIPint now allows individual vehicle manufacture specific payload types for the UUDT (Unacknowledged Unsegmented Data Transfer).

DEXT Export Event Port Mappings (MSR4-32363)

The DEXT export now also contains Event Port Mappings for Events, Operation-Cycles, Event- and Storage-Conditions and Indicators.

DEXT Export Port Mappings for FID (MSR4-32363)

The DEXT export now also contains port mappings for function identifiers (FID).

New API Dem_GetFreezeFramePrestored() (MSR4-32412)

The Dem module now provides a new API Dem_GetFreezeFramePrestored(). This API returns TRUE if a freeze frame is currently prestored for the event (prestored in temporary buffers, waiting to be persisted into memory) or FALSE if the prestored freeze frame is stored into memory or discarded (already processed).

64-Bit Support in DDM (MSR4-32632)

DDM now supports 64-bit. With this, CDD/ODX-Import can now handle huge input files.

Support of UDS Service 0x29 0x04 in Dcm (MSR4-32640)

The Dcm module now supports the UDS service 0x29 0x04 TransmitCertificate.

n/k+1 Socket Optional for DoIP (MSR4-33125)

ISO-13400 specifies to support a n/k+1 socket to reject additional tester connection attempts, where n/k is the number of concurrent TCP/TLS data connections supported by the DoIP entity. The +1 socket has now been made optional to reduce the memory consumption of the entire project.

DoIP Routing Activation on TCP To Detect Unused TLS Connections (MSR4-33126)

DoIP now performs an alive check on the TLS connection upon reception of the routing activation request on the TCP socket if the tester is already connected on the TLS connection. With this, TLS connections are closed if unused.

Hardware Security Module

Component-Based Folder Structure in veHsm Deliveries (MSR4-31324)

The folder structure of veHsm deliveries has been aligned with MICROSAR Classic deliveries and are now component-based.

Measurement and Calibration

Support C-API Access To Measurement Points in Rtm (MSR4-31201)

Rtm now supports the following C-API calls:

- Rtm_PrepareMPSettings (selects the measurement points to be active when measurement is started)
- Rtm_StartMeasurement (triggers a measurement start in next Rtm_MainFunction)
- Rtm_StopMeasurement (triggers a measurement stop in next Rtm_MainFunction)
- Rtm_ClearMeasurementResults (triggers clear of measurement results in next Rtm_MainFunction)
- Rtm_GetMeasurementItem (now supports all measurement points)

These API calls can be used instead of Xcp-based steering and reporting.

SoAd Burst Mode Supported in TcpIpXcp (MSR4-31615)

MICROSAR Classic XCP on TcpIp now supports SoAd burst mode for transmitting data. As a result, the transmission rate of Xcp measurements could be increased.

Memory

vMemAccM Available (-)

In MICROSAR Classic SIPs that contain the module Fee, the module vMemAccM (Memory Access Manager) is now also delivered which acts as an interface to the Vector memory driver veMem and coordinates the nonvolatile memory access by multiple users to avoid conflicts with shared resources. To use vMemAccM without veMem, wrappers are provided that enable the usage of the AUTOSAR Fls driver as veMem (vMem_30_Fls) and which add an AUTOSAR Fls API for vMemAccM (Fls_30_vMemAccM).



Documentation in BSW Package
Refer to the User Manual Memory Architecture located in the .\Doc\UserManuals folder of your MICROSAR Classic SIP to find out if this solution can be used in your project.

Add or Remove Driver Functionality Temporary Via vMemAccM (MSR4-28984)

The vMemAccM can replace a configured vMem driver via an alternate, for example

Memory

downloaded, vMem driver and uses it to access the NV RAM and finally switch back to the original configuration. Use case is to temporary add or remove driver functionality to access specific memory information or to deny to rewrite memory parts.

Parallel Memory Device Access Supported in veMem Solution (MSR4-29640)

The memory access manager (veMemAccM) sets up multiple memory driver (veMem) jobs in parallel to speed up the NV RAM access.

New Modul Fee FlexNor (Beta) (MSR4-29797, MSR4-29360, MSR4-29798)

The new Fee FlexNor is a Vector own development and provides flash management functionality compatible to the AUTOSAR Fee. The module is currently available as Beta and has some limited functionalities:

- Flash Bootloader mode is not supported (no operation with a reduced block configuration on the same partition).
- No support for updates of the payload size for existing blocks.
- Limited robustness (incomplete write and erase operations can’t be handled yet).



Documentation in BSW Package
Refer to the Readme_Fee.pdf located in the root folder of you MICROSAR Classic SIP for more information.

AUTOSAR Compatible Calculation of NvM CRC32 Only (MSR4-30456)

Up to now, the NvM also supported a compatibility mode for non-AUTOSAR compliant CRC32 calculation in addition to the calculation according to the AUTOSAR standard. This backwards compatibility is now no longer applicable, so that only the calculation as specified by AUTOSAR is supported.



This feature requires the migration of existing projects:

Projects using the compatibility mode of the NvM CRC32 calculation must migrate to the AUTOSAR-conform CRC32 calculation.



Additional Information
https://kb.vector.com/entry/1802/

Performance Optimization of vMem Solution (MSR4-31191)

The performance of the vMem solution as a flash driver has been optimized regarding the processing timing of their repective jobs in the needed MainFunction cycle. Depending on the job type and hardware a reduction of processing time up to -50% could now be achieved.

Operating System

Support Trusted Applications With Protection in the OS (MSR4-1354, MSR4-1343)

The Gen7 OS was extended by the AUTOSAR parameters OsTrusted and OsTrustedApplicationWithProtection. With those, the Applications can now be assigned a trusted flag in the OS which results in the following three behaviors:

1: OsTrusted = 0 -- Nontrusted -- MPU is always active

2: OsTrusted = 1 & OsTrustedApplicationWithProtection = 0 -- Trusted -- MPU not active

3: OsTrusted = 1 & OsTrustedApplicationWithProtection = 1 -- Trusted -- MPU is active

In case the OsTrustedApplicationWithProtection is set to true (last case) the RTE will handle it internally as if it would be a non-trusted Application.

Runtime Environment

New Software Development Workflow With SwCluC (Beta) (MSR4-30195)

AUTOSAR introduced in R20-11 the new Flexibility concept which introduces the definition and setup of Software Cluster Connections (SwCluC) in the Application and the BSW. This introduced a new workflow possibility to develop more independently the Application-parts from the Host-Integration Setup on an ECU between different involved parties. Vector provides a comprehensive solution with embedded code and tooling support.



Additional Information
For more information refer to our YouTube video here.

Functionality of Rte Add-On Let Extended (MSR4-30309)

The Rte Add-On Let (Logical Execution Time) now supports multiple instantiations, PR-ports, and IsEvent ExecutionOrderConstraints.

Moreover, the memory consumption and runtime of the fill flush routines was reduced.

OslocDataTypeRef Supported in OS and RTE (MSR4-30731)

The RTE and OS generators have been aligned so that now both evaluate the information of the IOC parameter OslocDataTypeRef.

Support of Explicit Un-Initialization of RTE Elements (MSR4-30843)

It is now possible to explicitly spare the initialization or RTE generated variables, when theses will be assigned to a MemSection later, which is initialized as a whole. The need to define an InitValue will be omitted in case an AddrMethod is referenced with an SectionInitializationPolicy set to:

> NO-INIT

Runtime Environment
> CLEARED > POWER-ON-CLEARED In case an InitValue is specified, nevertheless the initialization by the RTE will take place.
RTE and Analyzer Stub Generation Optimized (MSR4-30866) The internal data structures for the RTE and analyzer stub generation were summarized. With this, the performance could be improved.
SetAbsAlarm() for Cyclic Trigger Implementation (MSR4-31133) The Rte now provides a configuration option that alternatively SetAbsAlarm() can be used for cyclic trigger implementation in RteStart(). Currently, the Rte always uses SetRelAlarm() which leads to different activation time for runnables that are triggered by an integer multiple of each other (e.g. 10ms, 20ms and 100ms). The problem with the use of SetRelAlarm() is that between the different calls for different frequencies the time advances so that in the end the alarms expire at different OS tick times and so a 10ms and a 20ms runnable are not executed at the same tick time. With the use of SetAbsAlarm(), it can be achieved that all runnables which have integer multiple cycle times are executed in the same OS tick. This ensures that the execution order of runnables is exactly as defined in the task mapping.
MemMap Code Section per OS Partition (MSR4-31364) Up to now, the complete RTE code has been mapped into one central code section on the main BSW core. Now, code parts located in the Rte_SystemApplication_OsCore_Core<x>.c files are placed into a core individual RTE code section to allow the linker to map this code into the core local memory (if supported by the hardware). With this, the performance can be increased, since the local flash memory of a core that can be accessed much faster than a remote flash memory of another core.
Optimized Buffer for Autonomous Error Response for SomeIp (MSR4-31719) The SomeIp buffer size for error response has been reduced to 8 Bytes.
Queued S/R Communication of LdCom With Dynamic Length Data Types (MSR4-32104) Rte now supports queued sender-receiver communication of LdCom with dynamic length data types. Note that cross-partition communiation, TP API, and TriggerTransmit are not supported.
Header File as Type Emitter Supported (MSR4-32176) The Rte now supports header files as type emitter for implementation data types as specified in AUTOSAR R21-11.

Runtime Environment

Rte DID and Calibration Ports API: Support of Pointer to a Specific Type (MSR4-32368)

The non-base type passing prototypes in Rte_<Swc>.h have been readded to the Rte generator when the preprocessor define RTE_PTR2ARRAYTYPE_PASSING is set. With this, AUTOSAR3 SWC implementations can still be used in AUTOSAR4.

Direct Access to NV RAM Mirror Allowed in RTE (MSR4-32780)

The RTE has been extended to allow direct access to the NV RAM mirror as described in SWS_Rte_03878. In addition, the RTE also allows direct access for SWC that access the NV RAM data via a NV-S/R Port on a different ECUC partition than the NV-SWC. With this, the performance could be improved.

McSupportData Support in RTE Generator (RTE-5606, MSR4-30181)

Starting with R28, RTE generator now also supports McSupportData generation.

McSupportData is the measurement and calibration description that is specified in AUTOSAR and can be used to create A2L files with the DVMcDataConverter that is provided as part of DaVinci Configurator Classic.

Note
Davinci Configurator Classic 5.25 SP1 and McDataConverter 2.04 are required for the usage of the new A2L workflow with McSupportData.

This feature requires the migration of existing projects:

McSupportData is only generated when so called flat instance descriptors for the measurement or calibration elements are configured.

The solving action of the RTE1239 validation message can create the flat instance descriptors, if needed.

In case McSupportData shall be used, the Rte.a2l, Rte_MemSeg.a2l and Rte_XcpEvents.a2l files no longer need to be included in the Master.a2l file. The RTE elements are automatically added to the McData.a2l that is produced by DVMcDataConverter.

However, it is not required to use McSupportData. Instead, also the Rte.a2l files that are generated directly by the RTE generator can still be included in Master.a2l.

New features will only be added to McSupportData in the future, however.

Beware that also the handling for DVMcDataConverter changed in R28. The generated McData.a2l file is no longer only a snippet that needs to be included in the Master.a2l file. Instead, it is a complete a2l file with all relevant information that can be passed directly to other tools (e.g. ASAP2Updater).

Runtime Environment

The path to the input Master.a2l needs to be given in the project settings editor in DaVinci Configurator Classic.

Safety

KeyM Up To ASIL D Available (MSR4-29019, SEC-1877)

The KeyM is now available for Freedom from Interference up to ASIL D.



This feature requires the migration of existing projects:

The KeyM now uses a locking mechanism that ensures that only one of the following API functions is being executed at the same time by rejecting a simultaneously called request with the error code given in the brackets:

- KeyM_ServiceCertificate (error code: KEYM_E_BUSY)
- KeyM_ServiceCertificateById (error code: KEYM_E_BUSY)
- KeyM_SetCertificate (error code: E_NOT_OK)
- KeyM_SetCertificateWithConstPtr (error code: E_NOT_OK)
- KeyM_SetCertificateInGroup (error code: KEYM_E_BUSY)
- KeyM_GetCertificate (error code: E_NOT_OK)
- KeyM_CertElementGet (error code: E_NOT_OK)
- KeyM_CertElementGetFirst (error code: E_NOT_OK)
- KeyM_CertElementGetNext (error code: E_NOT_OK)
- KeyM_CertificateElementGetCount (error code: E_NOT_OK)
- KeyM_CertificateElementGetByIndex (error code: E_NOT_OK)
- KeyM_VerifyCertificates (error code: KEYM_E_BUSY)
- KeyM_VerifyCertificate (error code: KEYM_E_BUSY)
- KeyM_VerifyCertificateChain (error code: KEYM_E_BUSY)
- KeyM_VerifyCertificateChainWithConstPtr (error code: KEYM_E_BUSY)
- KeyM_VerifyGroup (error code: E_NOT_OK)
- KeyM_InitCSR (error code: E_NOT_OK)
- KeyM_SetCRE (error code: E_NOT_OK)
- KeyM_GetIssuerCertId (error code: E_NOT_OK)
- KeyM_GetGroupCertId (error code: E_NOT_OK)
- KeyM_CertStructureGet (error code: E_NOT_OK)
- KeyM_CsrElementSet (error code: E_NOT_OK)
- KeyM_GetCertHash (error code: KEYM_E_BUSY)

Safety

This locking mechanism also prevents that one and the same function is being executed repeatedly at the same time independently of its doxygen declaration for being reentrant or not and independently of the certificate in process.

Due to this new locking mechanism function calls, requested at the same time, e.g., from different tasks, might be rejected either with return value KEYM_E_BUSY or with return value E_NOT_OK. I.e., function calls that have been successfully processed so far, now might be rejected.

Thus, it must be ensured that the KeyM is only processing one request at a time, e.g., by calling all KeyM API functions from one and the same task.

Support E2E Profile 22 (MSR4-29530)

The E2E Profile 22 according to AUTOSAR 4.3.1 is now available

StbM Safety Use-Case Extended (MSR4-31414)

The functional safe scope of StbM has been extended by the following functionalities:

- Offset and rate correction for timebases.
- Usage of pure local time timebases.
- pDelay measurement.
- Rate correction.

Furthermore, the supported ASIL level of the functional safe scope of StbM has been increased from ASIL B to up to ASIL D.

E2ELib Adapted to the E2EProtocol R20-11 State Machine Specification (MWDG-5650, MSR4-31513)

The E2ELib now supports the extensions of the E2E State Machine as specified in AUTOSAR R20-11, listed in chapter 6.15 Specification of E2E state machine in AUTOSAR_PRS_E2EProtocol.



This feature requires the migration of existing projects:

Background:

The E2E state machine is used by calling E2E_SMCheckInit() and E2E_SMCheck() services. Both services have a parameter "const E2E_SMConfigType* ConfigPtr". The Config structure of type E2E_SMConfigType must be instantiated and defined by the User of the E2E State machine.

The E2E_SMConfigType is now provided by E2E_SM.h according to AUTOSAR E2E specification R20-11. This means the member TransitToInvalidExtended was added. For detailed explanation of the new member and its impact to state machine behavior see

Safety

the AUTOSAR E2E specification R20-11 (Note: TransitToInvalidExtended is erroneously not directly specified in [SWS_E2E_00342]).

Migration:

Setting the new member TransitToInvalidExtended to FALSE disables the new implemented transitions from state E2E_SM_NODATA and state E2E_SM_INIT to state E2E_SM_INVALID. The internal behavior of the state machine would not change in this case.

E2ELib Adapted to the E2EProtocol R19-11 State Machine Specification (MWDG-6638, MSR4-32167)

The E2ELib now supports the additional parameters for the E2E windows size, namely WindowSizeInit, WindowSizeInvalid, and WindowSizeValid replacing the singular WindowSize parameter as deifned in AUTOSAR < R19-11. This allows a context-sensitive switch of the window size based on whether recently received messages were valid.



This feature requires the migration of existing projects:

Background:

The E2E state machine is used by calling E2E_SMCheckInit() and E2E_SMCheck() services. Both services have a parameter "const E2E_SMConfigType* ConfigPtr". The Config structure of type E2E_SMConfigType must be instantiated and defined by the User of the E2E State machine.

The E2E_SMConfigType is now provided by E2E_SM.h according to AUTOSAR E2E specification R19-11. This means, its former member WindowSize was replaced by WindowSizeInit, WindowSizeInvalid, and WindowSizeValid. Additionally, ClearToInvalid was added. For detailed explanation of the new member see the AUTOSAR E2E specification R19-11.

Migration:

Setting the three new members WindowSizeInit, WindowSizeInvalid and WindowSizeValid to the former WindowSize value AND setting ClearToInvalid to FALSE gets the closest to the former internal behavior of the state machine. In this case, only the transition from state E2E_SM_INVALID to state E2E_SM_VALID is now stricter as every ERROR message will reset the OK message count.

Security

Core Driver Update to Restrict Write Access of SOK Keys from Host side (HSM-2031, MSR4-31832)

Security

Keys shall not be accessible through any API from Host side which could manipulate these keys (e.g. KeyElementSet/KeyDerive). Furthermore, the keys should be able to be used for crypto operations.



This feature requires the migration of existing projects:

To restrict the write access, the core driver have been updated and require the migration mentioned below.

VKMS Key as crypto keys in Crypto_30_VKMS_Core:

All VKMS Keys must be configured as crypto keys in Crypto_30_VKMS_Core as well. The original VKMS keys in the Crypto_30_LibCv must be defined as Hsm internal only keys to avoid direct access from Host side. The corresponding key references in the CryIf must be either changed to key located at the Crypto_30_VKMS_Core or new CryIf keys must be created.

Crypto job using VKMS Keys must be handled by Crypto_30_VKMS_Core:

Crypto_30_VKMS_Core provides all available driver objects usable with VKMS keys for Host side (Crypto_30_LibCv and Crypto_30_Hwa). The driver objects have the suffix "_VKMSKeyAccess". Every crypto job using VKMS Keys must be handled by the Crypto_30_VKMS_Core.

Safety checks in Crypto_30_LibCv must be enabled:

/MICROSAR/Crypto_30_LibCv/Crypto/CryptoGeneral/CryptoSafeBswChecks must be enabled to ensure correct key ids are used.

Crypto_30_VKMS_Core key ids are greater than 30000.



Documentation in BSW Package
Refer to the Technical Reference Crypto VKMS Core for more information.

Postbuild-Selectable and -Loadable Support in SokFm (MSR4-30861)

SokFm now supports Postbuild-Selectable and Postbuild-Loadable.

Multi-Processor Deployment of IdsM Supported (MSR4-31469)

IdsM now supports multi-processor deployment. With this, reporting of security events and filtering is supported for multiple micro-controllers and persisting of qualified security events (QSEv) and/or transmission of QSEvs to the IdsR is centralized on a host microcontroller. Note that the configuration of the IdsM instances must be done in separate DaVinci Configurator Classic projects.

MemMap Usage in vSecPrim (MSR4-31533, SEC-3669)

Security



This feature requires the migration of existing projects:

From now on, the MemMap sections can be generated into a file called vSecPrim_MemMap.h. This requires including the Common_MemMap generator into the project and using the file vSecPrim_MemMap.arxml to extract the used memory sections.

In order to keep compatibility to older projects, a file vSecPrim_MemMap_Compatibility.h is provided in the implementation folder. During the SWCP, this file is renamed to vSecPrim_MemMap.h and it relies on the file MemMap.h which was used until now.

Naming of Memory Sections in vIKE (MSR4-31533, SEC-3654)



This feature requires the migration of existing projects:

The naming of memory sections has changed from AUTOSAR 4.0.0 style to AUTOSAR 4.0.3 style. If memory mappings are used, they need to be migrated to the new section names.

Please migrate memory sections from

VIKE_<START|STOP>_SEC_<OLD>

to

VIKE_<START|STOP>_SEC_<NEW>

as follows:

- OLD: CONST_8BIT -> NEW: CONST_8
- OLD: CONST_16BIT -> NEW: CONST_16
- OLD: CONST_32BIT -> NEW: CONST_32
- REMOVED: CONST_64BIT
- UNCHANGED: CONST_UNSPECIFIED
- OLD: VAR_NOINIT_8BIT-> NEW: VAR_NO_INIT_8
- OLD: VAR_ZERO_INIT_8BIT -> NEW: VAR_CLEARED_8

Safety Requirements Check Using ECDSA in Crypto_30_LibCv and vSecPrim (MSR4-31543)

The Crypto_30_LibCv and vSecPrim modules now check the integrity of the signed content using ECDSA with ASIL B integrity.

Persisting of QSEv in Main Diagnostic Memory Supported by IdsM (MSR4-31612)

IdsM now supports to store qualified security events (QSEv) in the main diagnostic memory.

Security

Storage of Certificates of Different Curves in One Container in KeyM (MSR4-31664)

KeyM now allows the storage of X509v3 certificates in one container, independent of the curve that was used to create them. Up to now, separate containers had to be created for each curve and certificate and a RAM buffer was also required for each container. Now, KeyM treats a container as a storage unit for X509v3 certificates and when evaluating the content, it analyzes which curve was used to create it and then processes it accordingly. The certificate storage in shared certificate slots is only supported for certificates with storage type RAM.

KeyM Provides Detailed Result for Group Validation of Certificates (MSR4-31665)

KeyM introduced a new callback notification function KeyMCertificateVerifyGroupCallbackNotificationFunc which notifies the application when a group verification has finished and provides the overall group verification status.

By additionally configuring the callback notification function KeyMCertificateVerifyCallbackNotificationFunc per certificate, KeyM provides individual certificate validation status of each group member certificate.

It allows more detailed feedback to the user module why a validation of a group failed or was not parsed and which specific certificate was responsible for it.

Overwrite Signature Verification Results (MSR4-31681)

It is now possible to overwrite the results of the signature verification via SecOC_VerifyStatusOverride(). SokFm now provides a diagnostic routine control that uses the APIs SokFm_RoutineControl_FunctionDeactivation_StartRoutine and SokFm_RoutineControl_FunctionDeactivation_RequestResultsRoutine. The sub set of SecOCRxProcessings that are deactivated can be specified in the configuration.

Support of SHA3 Crypto Acceleration in veHsm (MSR4-31986)

The module veHsm now supports SHA3-256 as specified by AUTOSAR for dedicated Platforms supporting SHA3. Details can be found in the microcontroller-specific Product Information.

Support of ECC384 Crypto Acceleration in veHsm (MSR4-31987)

The module veHsm now supports ECC384 for dedicated Platforms supporting ECC384 accelerated in hardware.

Support of VKMS Version 2.8 (MSR4-32010)

The VKMS supports now the specification version 2.8.



This feature requires the migration of existing projects:

With the support of VKMS version 2.8, the VKMS has a fixed relationship to Rte_VKMS_

Security
Types.h. Therefore, if no Rte is used and thus Rte_VKMS_Types.h is not generated, an empty template for Rte_VKMS_Types.h must be created.
MemMap Usage in SecOC (MSR4-32079)  **This feature requires the migration of existing projects:** From now on, the SecOC MemMap sections can be generated into a file called SecOC_MemMap.h. This requires including the Common_MemMap generator into the project and using the file SecOC_MemMap.arxml to extract the used memory sections.
NIST SP 800-56C Option 1 Supported (MSR4-32244) The modules vSecPrim and LibCV now also support Key Derivation NIST SP 800-56C Option 1 in addition to the already supported Option 2.
Csm Optimization (MSR4-33004) Speed up retriggering of asynchronous job processing in case of empty queue in Csm.
Support of Further PSK Cipher Suites for TLS Server (MSR4-33221, MSR4-33789) The AUTOSAR TLS Server supports now additional PSK Cipher Suites such as TLS_PSK_WITH_AES_128_GCM_SHA256 and TLS_PSK_WITH_NULL_SHA256. The complete list can be found in the Technical Reference of the TCPIP module.
Reply To CERTREQ if No Matching CA Is Provided (MSR4-33923) If no certificate can be found based on the received CERTREQ, vIKE ignores the CERTREQ. Instead it transmits all available certificates that fit into a reply message, thus allowing the sender to validate the recipient of the CERTREQ with an alternate certificate.
ECBD Support in Crypto (Sw) (MSR4-949) Crypto (Sw) now supports ECBD (Elliptic Curve Burmester Desmedt) with SECp224r1 and NIST P-224.
Key Value Changed Callouts Supported in Crypto (SEC-3570, MSR4-31251)  **This feature requires the migration of existing projects:** Crypto_30_LibCv provides a new internal header file Crypto_30_LibCv_InternalApi.h. This must be taken into account when different make files are used than those included in the module.
MemMap Usage in Crypto LibCv (SEC-3640, MSR4-31533)

Security

This feature requires the migration of existing projects:

From now on, the MemMap sections can be generated into a file called Crypto_30_LibCv_MemMap.h. This requires including the Common_MemMap generator into the project and using the file Crypto_30_LibCv_MemMap.arxml to extract the used memory sections.

In order to keep compatibility to older projects, a file Crypto_30_LibCv_MemMap_Compatibility.h is provided in the implementation folder. During the SWCP, this file is renamed to Crypto_30_LibCv_MemMap.h and it relies on the file MemMap.h which was used until now.

In files which do not belong to the implementation subpackage of the module Crypto_30_LibCv and which use memory section of the module Crypto_30_LibCv, you must ensure to include the file Crypto_30_LibCv_MemMap.h instead of the file MemMap.h. This could e.g. be the case in header files for the Schedule Manager.

Software Download

Decompression for OTA Download (Vector) (MSR4-29267)

The vFotaH now supports a Vector decompression algorithm to reduce the transfer size of the software update data.

Tooling

Network License Support (DVCFG-6362)

Support of the NVLM-based Network Licenses, including Floating, Named User (Network Activation) and CI MICROSAR Based License support.

This replaces existing Floating and Automation Server License Types in FlexNet Technology.

Additional Information
https://support.vector.com/kb?id=kb_article_view&sysparm_article=KB0013664

Follow-up: Support for RTE Implementation Plug-in (DVCFG-6908)

Extended Support for RTE Implementation plug-in:

- Validation provided to check existence of referenced rtePluginProps at FlatInstanceDescriptors.
- Properties View provides information for new RIPS FlatInstanceDescriptor columns.

Tooling

- Completed State of Component Port Considering RIPS References adapted.

Relaunch: DaVinci Configurator Classic (DVCFG-6938)

The product DaVinci Configurator Pro is replaced by the new product DaVinci Configurator Classic, including the product options Runtime Environment (.RTE) , Workflow (.WF) and Module Development (.MD).

Old product DaVinci Developer is replaced by the new product DaVinci Developer Classic, including the product option Contract Phase Generation (.CPG).

Old product option DaVinci Configurator Pro Option .CI is replaced by the new product DaVinci Team SE.



Important Information
Usage of New DaVinci Products requires a new Product License.

All customers with active Old Product License (under maintenance or subscription) can exchange

the old Product License to a New Product License free of charge.

The license exchange is a manual process. Please get in contact with your Vector Tool Account Manager in case of urgent needs.



Additional Information
https://support.vector.com/kb?id=kb_article_view&sysparm_article=KB0013663

Support for Logical Execution Time (LET) Modelling Step2/R28 (DVCFG-7011)

Extended Support for Logical Execution Time (LET) Modelling:

- Introduced editor 'Logical Execution Time' in 'Timing' domain.
  - Overview LET frames 'LET Intervals', a grid is provided summarizing all defined LET frames.
  - Specific LET frame view.
- Validate Order of EOC Groups.
- Import external LET configuration via project directory folder.
- Execution Order Constraints: Changed Direct Successor Handling.
- Support EOCs with EOCExecutableEntityRefGroup and FlatExtract references.

Existing ComfortView Extended for CanIfTxPduCfgs (DVCFG-7272)

FullCan Assistant introduced to select controller and HW object and parameterizes the existing operations.

Tooling

Filter Issue Reports in DaVinci Package Manager (DVPM-1788)

With the new version 1.38.0 of DaVinci Package Manager, issue reports are now filtered based on the used DPA project. For this, new menu commands are provided in the Projects tab of the DaVinci Package Manager:

- Update Filtered Issue Report: Generates the filtered issue report.
- Show Filtered Issue Report: Opens the generated issue report.

In addition, a script/command line-based approach is also supported. To filter an issue report via the command line interface, run the following command:

filter-issue-report <dpa-file-path> <issue-report-path> <filter-issue-output>

- <dpa-file-path> : (Required) Path to a DPA file.
- <issue-report-path> : (Required) Path to an issue report file. (Can also be a folder of Issue Reports)
- <filter-issue-output> : (Required) Path where the filtered issue report will be created.

This will generate a filtered issue report in the output directory using the DPA project.

Currently, not all issues can be filtered, but more and more issues will be updated to support filtering so that the filtering will be continuously improved.

Note that a Davinci Configurator Classic license is required.

Please contact Vector for more information or in case a license is required.

Preparation for Enhanced BSW Initialization for Multi Partition/Core Configurations (MSR4-177)

For this feature, the initialization information of the BSW modules will be moved from the module EcuC to the module vSet. Temporary, the initialization information is stored in both modules, vSet and EcuC, but currently only the information in EcuC is used for initialization.



This feature requires the migration of existing projects:

To prevent validation errors regarding initialization service, it is necessary to activate the Postbuild-Selectable Support in the module settings of the module vSet in the DaVinci Configurator Classic.

Migration Steps

Charge

veScc Uses KeyM for Certificate Handling & AUTOSAR TLS Client instead of vTLS Client (MSR4-29622, MSR4-29623)

As a result, veScc will only perform the certificate handling specified by the ISO15118, all additional handling e.g. for OEM-specific update mechanisms must now be implemented by the customer application using the KeyM module directly

➡ Feature description: **veScc Uses KeyM for Certificate Handling & AUTOSAR TLS Client instead of vTLS Client (MSR4-29622, MSR4-29623)**

Communication

Com_IpduGroupControl Removed (COM-2458, MSR4-31460)

For the migration of existing projects, a migration script is provided.

➡ Feature description: **Com_IpduGroupControl Removed (COM-2458, MSR4-31460)**

ComDescriptionRoutingCodeGeneration Removed (COM-2523)

For the migration of existing projects, a script is delivered.

➡ Feature description: **ComDescriptionRoutingCodeGeneration Removed (COM-2523)**

Channel-specific LinIf Main Functions (LIN-508)

The previous LinIf_MainFunction does not exist anymore, but is replaced by 1 to n LinIf_MainFunction_<channelname> main functions.

Each of these new LinIf main functions must be mapped to an OS task again.

➡ Feature description: **Channel-specific LinIf Main Functions (LIN-508)**

Option To Disable IPv4 UDP Checksum Calculation in TcpIp (MSR4-32103, TCPIP-4355)

With the introduction of this AUTOSAR feature, the handling of received/transmitted IPv4 UDP frames with checksum == 0 is changed: Received IPv4 UDP frames with checksum == 0 will be discarded by the TcpIp implementation and not forwarded to the upper layer.

To allow such frames, the functionality must be unlocked per socket:

> For SoAd as upper layer by disabling parameter:

/SoAd/SoAdConfig/SoAdSocketConnectionGroup/SoAdSocketProtocol/SoAdSocketUdp/SoAdSocketUdpChecksumEnabled

Communication

> For customer applications as upper layer by usage of TcpIp_ChangeParameter() API.

If the functionality is unlocked, the TcpIp module will also use checksum == 0 for transmission of IPv4 UDP packets on the corresponding socket(s).

See AUTOSAR requirements SWS_TCPIP_00279 and SWS_TCPIP_00280 and technical documentation for further details.

➡ Feature description: **Option To Disable IPv4 UDP Checksum Calculation in TcpIp (MSR4-32103, TCPIP-4355)**

MemMap Usage in E2ELib (MWDG-5808, MSR4-31533)

Background:

The E2ELib now uses the dynamically generated E2E_MemMap.h for the memory mapping, to reduce the compile time. E2E Library is non-generated, deterministic software code, where all inputs and settings are passed by function parameters. So in previous versions it was not possible to add it to the configuration. But now it provides a dummy BSWMD, which must be added to the configuration, that the MemMap generator recognizes it and generates the E2E_MemMap.h.

Migration:

To generate the E2E_MemMap.h the E2ELib must be added to the configuration manually.

Compatibility:

To be compatible with previous releases (before R27) the E2ELib still provides the E2E_MemMap.inc. In this case the memory sections are mapped via the MemMap.h which is included by a static E2E_MemMap.h. No migration is necessary.

➡ Feature description: **MemMap Usage in E2ELib (MWDG-5808, MSR4-31533)**

Diagnostics

First Part of Supporting User-Defined Fault Memories According To ISO 14229-1 - Use-Case SEM (DEM-6320, MSR4-1230)

It will no longer be possible to map internal events (events with no DTC reference) to primary/user-defined memories anymore. Hence, it will not be supported to clear internal events from primary and user-defined memories separately anymore.

For the purpose of clearing the events, all internal events will be virtually mapped to the event destination referenced by DemGeneral/DemClearEventsWithoutDTCEventMemoryRef. They can be cleared only

Diagnostics

by clearing the event destination referenced by DemClearEventsWithoutDTCEventMemoryRef.

While upgrading a configuration with internal events to this software revision, the Dem will add a default value for DemGeneral/DemClearEventsWithoutDTCEventMemoryRef. This value must be verified for its compatibility with your requirements.

➡ Feature description: **First Part of Supporting User-Defined Fault Memories According To ISO 14229-1 - Use-Case SEM (DEM-6320, MSR4-1230)**

MemMap Usage in vDam (HSM-2208, MSR4-31533)

From now on, the MemMap sections can be generated into a file called vDam_MemMap.h. This requires including the Common_MemMap generator into the project and using the file vDam_MemMap.arxml to extract the used memory sections.

In order to keep compatibility to older projects, a file vDam_MemMap_Compatibility.h is provided in the implementation folder. During the SWCP, this file is renamed to vDam_MemMap.h and it relies on the file MemMap.h which was used until now.

➡ Feature description: **MemMap Usage in vDam (HSM-2208, MSR4-31533)**

vDiagSecAcc module specific MemMap support (MSR4-31533, SEC-3631)

From now on, the MemMap sections can be generated into a file called vDiagSecAcc_MemMap.h. This requires including the Common_MemMap generator into the project and using the file vDiagSecAcc_MemMap.arxml to extract the used memory sections.

In order to keep compatibility to older projects, a file vDiagSecAcc_MemMap_Compatibility.h is provided in the implementation folder. During the SWCP, this file is renamed to vDiagSecAcc_MemMap.h and it relies on the file MemMap.h which was used until now.

➡ Feature description: **vDiagSecAcc module specific MemMap support (MSR4-31533, SEC-3631)**

Memory

AUTOSAR Compatible Calculation of NvM CRC32 Only (MSR4-30456)

Projects using the compatibility mode of the NvM CRC32 calculation must migrate to the AUTOSAR-conform CRC32 calculation.

Additional Information
https://kb.vector.com/entry/1802/

➡ Feature description: **AUTOSAR Compatible Calculation of NvM CRC32 Only**

Memory

(MSR4-30456)

Runtime Environment

McSupportData Support in RTE Generator (RTE-5606, MSR4-30181)

McSupportData is only generated when so called flat instance descriptors for the measurement or calibration elements are configured.

The solving action of the RTE1239 validation message can create the flat instance descriptors, if needed.

In case McSupportData shall be used, the Rte.a2l, Rte_MemSeg.a2l and Rte_XcpEvents.a2l files no longer need to be included in the Master.a2l file. The RTE elements are automatically added to the McData.a2l that is produced by DVMcDataConverter.

However, it is not required to use McSupportData. Instead, also the Rte.a2l files that are generated directly by the RTE generator can still be included in Master.a2l.

New features will only be added to McSupportData in the future, however.

Beware that also the handling for DVMcDataConverter changed in R28. The generated McData.a2l file is no longer only a snippet that needs to be included in the Master.a2l file. Instead, it is a complete a2l file with all relevant information that can be passed directly to other tools (e.g. ASAP2Updater).

The path to the input Master.a2l needs to be given in the project settings editor in DaVinci Configurator Classic.

➡ Feature description: **McSupportData Support in RTE Generator (RTE-5606, MSR4-30181)**

Safety

KeyM Up To ASIL D Available (MSR4-29019, SEC-1877)

The KeyM now uses a locking mechanism that ensures that only one of the following API functions is being executed at the same time by rejecting a simultaneously called request with the error code given in the brackets:

- KeyM_ServiceCertificate (error code: KEYM_E_BUSY)
- KeyM_ServiceCertificateById (error code: KEYM_E_BUSY)
- KeyM_SetCertificate (error code: E_NOT_OK)
- KeyM_SetCertificateWithConstPtr (error code: E_NOT_OK)

Safety

- KeyM_SetCertificateInGroup (error code: KEYM_E_BUSY)
- KeyM_GetCertificate (error code: E_NOT_OK)
- KeyM_CertElementGet (error code: E_NOT_OK)
- KeyM_CertElementGetFirst (error code: E_NOT_OK)
- KeyM_CertElementGetNext (error code: E_NOT_OK)
- KeyM_CertificateElementGetCount (error code: E_NOT_OK)
- KeyM_CertificateElementGetByIndex (error code: E_NOT_OK)
- KeyM_VerifyCertificates (error code: KEYM_E_BUSY)
- KeyM_VerifyCertificate (error code: KEYM_E_BUSY)
- KeyM_VerifyCertificateChain (error code: KEYM_E_BUSY)
- KeyM_VerifyCertificateChainWithConstPtr (error code: KEYM_E_BUSY)
- KeyM_VerifyGroup (error code: E_NOT_OK)
- KeyM_InitCSR (error code: E_NOT_OK)
- KeyM_SetCRE (error code: E_NOT_OK)
- KeyM_GetIssuerCertId (error code: E_NOT_OK)
- KeyM_GetGroupCertId (error code: E_NOT_OK)
- KeyM_CertStructureGet (error code: E_NOT_OK)
- KeyM_CsrElementSet (error code: E_NOT_OK)
- KeyM_GetCertHash (error code: KEYM_E_BUSY)

This locking mechanism also prevents that one and the same function is being executed repeatedly at the same time independently of its doxygen declaration for being reentrant or not and independently of the certificate in process.

Due to this new locking mechanism function calls, requested at the same time, e.g., from different tasks, might be rejected either with return value KEYM_E_BUSY or with return value E_NOT_OK. I.e., function calls that have been successfully processed so far, now might be rejected.

Thus, it must be ensured that the KeyM is only processing one request at a time, e.g., by calling all KeyM API functions from one and the same task.

Feature description: **KeyM Up To ASIL D Available (MSR4-29019, SEC-1877)**

E2ELib Adapted to the E2EProtocol R20-11 State Machine Specification (MWDG-5650, MSR4-31513)

Background:

The E2E state machine is used by calling E2E_SMCheckInit() and E2E_SMCheck() services. Both services have a parameter "const E2E_SMConfigType* ConfigPtr". The Config structure of type E2E_SMConfigType must be instantiated and defined by the

Safety

User of the E2E State machine.

The E2E_SMConfigType is now provided by E2E_SM.h according to AUTOSAR E2E specification R20-11. This means the member TransitToInvalidExtended was added. For detailed explanation of the new member and its impact to state machine behavior see the AUTOSAR E2E specification R20-11 (Note: TransitToInvalidExtended is erroneously not directly specified in [SWS_E2E_00342]).

Migration:

Setting the new member TransitToInvalidExtended to FALSE disables the new implemented transitions from state E2E_SM_NODATA and state E2E_SM_INIT to state E2E_SM_INVALID. The internal behavior of the state machine would not change in this case.

➡ Feature description: **E2ELib Adapted to the E2EProtocol R20-11 State Machine Specification (MWDG-5650, MSR4-31513)**

E2ELib Adapted to the E2EProtocol R19-11 State Machine Specification (MWDG-6638, MSR4-32167)

Background:

The E2E state machine is used by calling E2E_SMCheckInit() and E2E_SMCheck() services. Both services have a parameter "const E2E_SMConfigType* ConfigPtr". The Config structure of type E2E_SMConfigType must be instantiated and defined by the User of the E2E State machine.

The E2E_SMConfigType is now provided by E2E_SM.h according to AUTOSAR E2E specification R19-11. This means, its former member WindowSize was replaced by WindowSizeInit, WindowSizeInvalid, and WindowSizeValid. Additionally, ClearToInvalid was added. For detailed explanation of the new member see the AUTOSAR E2E specification R19-11.

Migration:

Setting the three new members WindowSizeInit, WindowSizeInvalid and WindowSizeValid to the former WindowSize value AND setting ClearToInvalid to FALSE gets the closest to the former internal behavior of the state machine. In this case, only the transition from state E2E_SM_INVALID to state E2E_SM_VALID is now stricter as every ERROR message will reset the OK message count.

➡ Feature description: **E2ELib Adapted to the E2EProtocol R19-11 State Machine Specification (MWDG-6638, MSR4-32167)**

Security

Core Driver Update to Restrict Write Access of SOK Keys from Host side (HSM-2031, MSR4-31832)

To restrict the write access, the core driver have been updated and require the migration mentioned below.

VKMS Key as crypto keys in Crypto_30_VKMS_Core:

All VKMS Keys must be configured as crypto keys in Crypto_30_VKMS_Core as well. The original VKMS keys in the Crypto_30_LibCv must be defined as Hsm internal only keys to avoid direct access from Host side. The corresponding key references in the CryIf must be either changed to key located at the Crypto_30_VKMS_Core or new CryIf keys must be created.

Crypto job using VKMS Keys must be handled by Crypto_30_VKMS_Core:

Crypto_30_VKMS_Core provides all available driver objects usable with VKMS keys for Host side (Crypto_30_LibCv and Crypto_30_Hwa). The driver objects have the suffix "_VKMSKeyAccess". Every crypto job using VKMS Keys must be handled by the Crypto_30_VKMS_Core.

Safety checks in Crypto_30_LibCv must be enabled:

/MICROSAR/Crypto_30_LibCv/Crypto/CryptoGeneral/CryptoSafeBswChecks must be enabled to ensure correct key ids are used.

Crypto_30_VKMS_Core key ids are greater than 30000.

Documentation in BSW Package
Refer to the Technical Reference Crypto VKMS Core for more information.

Feature description: **Core Driver Update to Restrict Write Access of SOK Keys from Host side (HSM-2031, MSR4-31832)**

MemMap Usage in vSecPrim (MSR4-31533, SEC-3669)

From now on, the MemMap sections can be generated into a file called vSecPrim_MemMap.h. This requires including the Common_MemMap generator into the project and using the file vSecPrim_MemMap.arxml to extract the used memory sections.

In order to keep compatibility to older projects, a file vSecPrim_MemMap_Compatibility.h is provided in the implementation folder. During the SWCP, this file is renamed to vSecPrim_MemMap.h and it relies on the file MemMap.h which was used until now.

Feature description: **MemMap Usage in vSecPrim (MSR4-31533, SEC-3669)**

Naming of Memory Sections in vIKE (MSR4-31533, SEC-3654)

The naming of memory sections has changed from AUTOSAR 4.0.0 style to AUTOSAR

Security

4.0.3 style. If memory mappings are used, they need to be migrated to the new section names.

Please migrate memory sections from

VIKE_<START|STOP>_SEC_<OLD>

to

VIKE_<START|STOP>_SEC_<NEW>

as follows:

- OLD: CONST_8BIT -> NEW: CONST_8
- OLD: CONST_16BIT -> NEW: CONST_16
- OLD: CONST_32BIT -> NEW: CONST_32
- REMOVED: CONST_64BIT
- UNCHANGED: CONST_UNSPECIFIED
- OLD: VAR_NOINIT_8BIT-> NEW: VAR_NO_INIT_8
- OLD: VAR_ZERO_INIT_8BIT -> NEW: VAR_CLEARED_8

Feature description: **Naming of Memory Sections in vIKE (MSR4-31533, SEC-3654)**

Support of VKMS Version 2.8 (MSR4-32010)

With the support of VKMS version 2.8, the VKMS has a fixed relationship to Rte_VKMS_Types.h. Therefore, if no Rte is used and thus Rte_VKMS_Types.h is not generated, an empty template for Rte_VKMS_Types.h must be created.

Feature description: **Support of VKMS Version 2.8 (MSR4-32010)**

MemMap Usage in SecOC (MSR4-32079)

From now on, the SecOC MemMap sections can be generated into a file called SecOC_MemMap.h. This requires including the Common_MemMap generator into the project and using the file SecOC_MemMap.arxml to extract the used memory sections.

Feature description: **MemMap Usage in SecOC (MSR4-32079)**

Key Value Changed Callouts Supported in Crypto (SEC-3570, MSR4-31251)

Crypto_30_LibCv provides a new internal header file Crypto_30_LibCv_InternalApi.h.

This must be taken into account when different make files are used than those included in the module.

Feature description: **Key Value Changed Callouts Supported in Crypto (SEC-3570, MSR4-31251)**

Security

MemMap Usage in Crypto LibCv (SEC-3640, MSR4-31533)

From now on, the MemMap sections can be generated into a file called Crypto_30_LibCv_MemMap.h. This requires including the Common_MemMap generator into the project and using the file Crypto_30_LibCv_MemMap.arxml to extract the used memory sections.

In order to keep compatibility to older projects, a file Crypto_30_LibCv_MemMap_Compatibility.h is provided in the implementation folder. During the SWCP, this file is renamed to Crypto_30_LibCv_MemMap.h and it relies on the file MemMap.h which was used until now.

In files which do not belong to the implementation subpackage of the module Crypto_30_LibCv and which use memory section of the module Crypto_30_LibCv, you must ensure to include the file Crypto_30_LibCv_MemMap.h instead of the file MemMap.h. This could e.g. be the case in header files for the Schedule Manager.

Feature description: **MemMap Usage in Crypto LibCv (SEC-3640, MSR4-31533)**

Tooling

Preparation for Enhanced BSW Initialization for Multi Partition/Core Configurations (MSR4-177)

To prevent validation errors regarding initialization service, it is necessary to activate the Postbuild-Selectable Support in the module settings of the module vSet in the DaVinci Configurator Classic.

Feature description: **Preparation for Enhanced BSW Initialization for Multi Partition/Core Configurations (MSR4-177)**

Release 27 (2021-09-24)

Implemented Features (incl. migration steps)

General

MemMap Generator Available And Module-specific MemMap Includes (MSR4-265, MSR4-31200, MSR4-30113)

The new module MemMap is now available. The module generates dedicated memory mapping header files for BSW modules as well as for software components as specified in AUTOSAR 19-11. The new module MemMap is now available. The module generates dedicated memory mapping header files for BSW modules as well as for software components as specified in AUTOSAR 19-11. With this, the compile time is improved through minimized header include dependencies and the configuration is centralized in the MemMap component. In addition, abstract configurations are provided within each component (description) that allow for generation of compiler-specific pragmas using vLinkGen.

For some modules, a module-specific <Mip>_MemMap.h must be configured in DaVinci Configurator Pro and generated via the MemMap generator. The following modules will use module-specific MemMaps:

- CAN: CanIf.
- CDD modules.
- CHARGE: vCanCcCdm.
- COM: IpduM, PduR.
- DIAG: vDrm.
- ETH: SoAd, EthTSyn.
- MEM: NvM.
- SYS: Det.
- RTE: Rte.

These modules will not be included in the general MemMap.h anymore.

This feature requires the migration of existing projects:

Custom-specific adaptions in the MemMap.h for these modules must be removed from the general MemMap.h and configured in DaVinci Configurator Pro.

Note
Refer to the Vector KnowledgeBase article for more information: https://support.vector.com/kb?id=kb_article_view&sysparm_article=KB0013394 .

General

Dynamic MemMap Generator (MSR4-31225)

The module MemMap now also supports components with dynamic MemMap sections.

AUTOSAR Monitoring and Debugging

vRtm Measurement Point Generation for RTE VFB Hooks (MSR4-1172)

The vRtm now also supports start and return hooks of Schedulable Entities (BSW Mainfunctions) for Runnable Measurement Points.

Rtm Extended for Monitoring Series ECUs (Beta) (MSR4-30652)

The Rtm now provides the following metric features:

- CPU Load with value percentile (histogram). Results are stored in the NvM during shutdown period.
- Task Response Times with value percentile (histogram). Results are stored in the NvM during shutdown period.
- Task Stack Consumption with value max. Results are stored in the NvM during shutdown period.

These features are available as Beta.

Rtm Extended for Monitoring Series ECUs (QM) (MSR4-30890)

The Rtm now provides the following metric features:

- CPU Load with value percentile (histogram). Results are stored in the NvM during shutdown period.
- Task Response Times with value percentile (histogram). Results are stored in the NvM during shutdown period.
- Task Stack Consumption with value max. Results are stored in the NvM during shutdown period.

These features are available as Production QM.

Charge

Support of Indirect CSM Access (MSR4-1366)

It is now possible to change the configuration of the modules vTls, vScc, and XmlSecurity so that they no longer directly call the CSM API. Instead, a different function name can be inserted which will then provide the calculation of the cryptographic routines for those modules.

Support of Bidirectional Power Transfer in vScc (Beta) (MSR4-1447)

Charge

vScc now supports Bidirectional Power Transfer (BPT) according to ISO/FDIS 15118-20 as Beta.



Additional Information
Refer to the following video for more information.

Support AC & DC According to ISO15118-20 in vScc (Beta) (MSR4-29598)

The vScc now supports AC & DC according to ISO/FDIS 15118-20 in dynamic and scheduled mode using EIM and TLS 1.2 as Beta.

CHAdeMO v3.0 Support in vCanCcCdm (MSR4-30270)

vCanCcCdm now supports CHAdeMO v3.0.

Support of HPC1 Service in vScc (MSR4-30824)

The vScc now supports the detection of HPC1 service with ID 63000 as specified in CharIN Guide ISO 15118 Service HPC1. In case an HPC1 service is detected, the renegotiation is disabled and MaxEntriesSAScheduleTuple is not sent.

Communication

Code Improvement in vBsdAd (-)

The vBsdAd code has been further improved so that all code and buffers specific to TCP are disabled when TCP is configured as OFF.

Configuration of Multiple MainFunctions in Com (COM-2215)

With the version 22.02.02 of the Com, the configuration of multiple MainFunctions is possible.



This feature requires the migration of existing projects:

With the support of the configuration of multiple MainFunctions, the names of the MainFunctions change and the MainFunction to Task Mapping is lost. If you want to preserve the MainFunction to Task Mapping, you must do the following steps for the migration of your configuration:

> Do the migration of your configuration.
> Run the script task RestoreReferencesRTEEVentToTaskMappings before you update the BswInternalBehavior of the Com.

Support of Indirect CSM Access (MSR4-1366)

It is now possible to change the configuration of the modules vTls, vScc, and

Communication

XmlSecurity so that they no longer directly call the CSM API. Instead, a different function name can be inserted which will then provide the calculation of the cryptographic routines for those modules.

MultiPartition Support in ComM and Nmlf (MSR4-150, MSR4-31259)

The ComM and Nmlf now support that one or more communication channels are mapped to a different partition than the main BSW partition, e.g. one channel on one partition and another channel on a different partition. The partitions can also be located on different cores.

TLS Server Available (Beta) (MSR4-168)

The TcpIp Add-On Tls now provides a TLS Server according to AUTOSAR 4.4. The TLS Server supports as follows:

- cipher suites:
  - TLS_ECDHE_ECDSA_WITH_NULL_SHA { 0xC0, 0x06 }
  - TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256 { 0xC0, 0x23 }
  - TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 { 0xC0, 0x2B }
  - TLS_ECDH_ECDSA_WITH_NULL_SHA { 0xC0, 0x01 }
  - TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256 { 0xC0, 0x25 }
  - TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256 { 0xC0, 0x2D }
- Elliptic Curves:
  - secp256r1 {23}
  - x25519 {29}

These features are available as Beta.

Postbuild-Selectable for Sd (MSR4-241)

The Sd now supports Postbuild-Selectable for the parameters SdClientServiceInstanceId and SdServerServiceInstanceId.

Support for DoIP Extension CONC_649 (MSR4-29025)

DoIP now supports the extension CONC_649_DoIPExtension according to AUTOSAR 19-11.



This feature requires the migration of existing projects:

Due to introduction of a new DoIPInterface container some manual corrections have to be applied when migrating a configuration created with prior MICROSAR Releases and performing Input file Updates:

1) Open configuration in MSR4-R27 SIP and execute SIP migration. There will be a new

Communication

container DoIP/DoIPConfigSet/DoIPInterface.

2) Perform Update of Input Files as suggested by DaVinci tool.

3) Merge content of the DoIPInterface containiner mentioned in 1) with the derived container.

4) Delete the DoIPInterface container mentioned in 1).

Optimization of DoIP Configuration (MSR4-29051)

DoIP has been optimized to ease the configuration of DoIP in general:

- Allow to reference the same DoIPRoutingActivation container by different DoIPTester (before: each DoIPTester required an own DoIPRoutingActivation)
- Allow to reference the same DoIPTargetAddress container by different DoIPChannel (before: each DoIPChannel required an own DoIPTargetAddress)

Note: This does not optimize the memory usage.

SoAd and Sd Measurement Data Supported (MSR4-30272)

The modules SoAd and Sd now support the following features:

- SoAd: Count of received invalid SOME/IP messages service ID.
- SoAd: Count of received invalid SOME/IP messages method ID.
- Sd: Count of received invalid SOME/IP SD messages.
- Sd: Count of received invalid SOME/IP subscriptions.
- SoAd: Reset count of received invalid SOME/IP messages service ID.
- SoAd: Reset count of received invalid SOME/IP messages method ID.
- Sd: Reset count of received invalid SOME/IP SD messages.
- Sd: Reset count of received invalid SOME/IP subscriptions.

For this, the Sd now provides the new API <Msn>_GetAndResetMeasurementData() as already available in the SoAd. The feature can be enabled and disabled via the configuration parameter SdGeneral/SdGetAndResetMeasurementDataApi and SoAdGeneral/SoAdGetAndResetMeasurementDataSomeIpSd.

Advanced Multi-Core Distribution for Ethernet (MSR4-30281)

The MICROSAR Ethernet stack now supports multi-core distribution on VLAN-level for specific controllers to enhance load balancing.



Additional Information
Refer to the following video for more information.

Communication

Support of Polling in Lin (MSR4-30285)

The Lin driver now supports polling. This allows a better support of the Flash Bootloader.

Multipartition and Multi-Core support in Com and LdCom (MSR4-30322, MSR4-30356)

Com and LdCom have been extended to support Multipartition and Multi-Core use-cases.



Important Information
The usage of the Signal Gateway in a Master-Satellite setup is currently not supported.

Multi-Partition Support in IpduM (MSR4-30331)

The IpduM now supports multi-partition. For this, the IpduM provides new Main Function APIs which must be called cyclically on the actual partition.

IDM Support in FrArTp (MSR4-30354, MSR4-29117)

The FrArTp now supports Postbuild-Selectable for IDM (FR).

SoAd Reporting of Security Events to IdsM (MSR4-30402)

SoAd now supports the reporting of security events to IdsM according to AUTOSAR 20-11.

Error Handling for Child SA Establishment in vIKE (MSR4-30457)

The error handling of vIKE for

- vIKE as a responder and receiving an IKE_AUTH request without matching child SA,
- vIKE as an initiator and receiving an IKE_AUTH response with error code NO_PROPOSAL_CHOSEN and
- vIKE replying to CREATE_CHILD_SA requests with error code NO_ADDITIONAL_SAS (both as Initiator and Responder)

has been aligned with RFC 7296.

Support of Version Driven Find Behavior in Sd (MSR4-30467)

The Sd now supports the version driven find behavior as specified in AUTOSAR R19-11. Therefore, the configuration parameter SdVersionDrivenFindBehavior can be set to EXACT_OR_ANY_MINOR_VERSION or MINIMUM_MINOR_VERSION to specify which minor versions are accepted for a client service when OfferService entries are received.

New Modul SoAdBsdAdapter for Posix (MSR4-30548)

Communication

The new module vBsdAd (SoAdBsdAdapter) for Posix is now available. The module enables MICROSAR Classic Ethernet communication on Linux, QNX and Integrity platforms as well as allows a customer implemented adaption. The functionality used to be contained in SoAd but has been redesigned into an own component.

AUTOSAR TLS Client Available (MSR4-30807)

The TcpIp Add-On Tls now provides a TLS Client for PSK handling according to AUTOSAR R19-11.

J1939-22 Supported (MSR4-31195)

J1939-22 (J1939 using CanFD) is now supported excluding J1939Tp extensions and placing of J1939Nm NM and CA PDUs in containers. Also, J1939Rm ACKM PDUs can only be placed in FEFF containers (i.e. using 29bit extended CAN ID).

CAN-Stack Supports Third Party CanDrv According to AUTOSAR 4.4.0 (MSR4-31234)

The MICROSAR CAN-stack was extended to support now third party MCAL CanDrv (e.g. from MCAL Vendors) according to the AUTOSAR version 4.4.0.

Streaming-Based Transmission Via TCP (MSR4-31329)

TcpIp now supports streaming-based transmission of data over TCP as specified in AUTOSAR. This enables RAM optimization for the TCP buffer and performance optimization for the transmission, especially for gateway routing use cases.

Disabling of vBsdAd for Safe Partitions (MSR4-31348)

The vBsdAd module can now be used in safety partitions and can there be safely disabled to ensure freedom from interference when used in series.



Important Information
The actual functionality of vBsdAd module has not yet been qualified for the necessary ASIL Safety level to be active and run in series projects. Please see Safety manual and TechRef for more information.

SdInstanceLocalAdressCheckLength in Sd Supported (MSR4-31429)

The Sd now accepts Offer-Service messages from endpoints located in a different subnet. Therefore, the parameter SdInstanceLocalAdressCheckLength is now supported and can be configured in SdInstance containers

TcpIp Add-On IPsec and Csm Interaction Optimized (MSR4-31576)

The performance of the TcpIp and Csm interaction in the IPsec use case has been improved. This results in speed advantages, especially in combination with HSM.

Communication

UDP Reception Speed for vBsdAd Increased (MSR4-31582)

The vBsdAd now supports the reception of multiple frames per UDP socket per main function cycle. The number of handled frames can be configured at configuration time. This allows to balance the runtime of the main function.

Local Address Management of vBsdAd Improved (MSR4-31583)

vBsdAd uses the POSIX API functions, getifaddrs and freeifaddrs to fetch system network interface information. Depending on the OS implementation, these function calls might take long time to execute and thus cause performance issues. With this feature the use of those calls has been optimized/lowered and it has been enabled for the use to configure a custom callout which is more optimized for the use case.

Option To Disable IPv4 UDP Checksum Calculation in TcpIp (MSR4-32103, TCPIP-4355)

TcpIp now supports disabling of IPv4 UDP checksum calculation in software at runtime. For this, the TcpIp_ChangeParameter() API has been extended by the option TCPIP_PARAMID_UDP_CHECKSUM as specified by AUTOSAR 4.3.0.

In case the UDP checksum calculation is disabled, the checksum of a transmitted frame is set to zero and received frames with checksum zero are not dropped but forwarded to the upper layer.



This feature requires the migration of existing projects:

With the introduction of this AUTOSAR feature, the handling of received/transmitted IPv4 UDP frames with checksum == 0 is changed: Received IPv4 UDP frames with checksum == 0 will be discarded by the TcpIp implementation and not forwarded to the upper layer.

To allow such frames, the functionality must be unlocked per socket:

> For SoAd as upper layer by disabling parameter:

/SoAd/SoAdConfig/SoAdSocketConnectionGroup/SoAdSocketProtocol/SoAdSocketUdp/SoAdSocketUdpChecksumEnabled

> For customer applications as upper layer by usage of TcpIp_ChangeParameter() API.

If the functionality is unlocked, the TcpIp module will also use checksum == 0 for transmission of IPv4 UDP packets on the corresponding socket(s).

See AUTOSAR requirements SWS_TCPIP_00279 and SWS_TCPIP_00280 and technical documentation for further details.

J1939Nm Available up to ASIL D (MSR4-324)

Communication

The module J1939Nm (excluding dynamic NM and gateway support) is now available for SafeBSW up to ASIL D.

Extended Support of Dynamic Length Frames in FrIf (SWAT-1323)

Frames configured with dynamic length can now contain any number of PDUs.

Diagnostics

Dcm Support of All Routine Data Types (MSR4-1205)

Dcm now also supports array and boolean data types for routines according to AUTOSAR.

Support of Dynamic vDrm Configuration (MSR4-225)

vDrm now supports dynamic CanTp connections, in which the CAN IDs to be used for transmission and reception are provided by the application.

Dcm Reporting of Security Events to IdsM (MSR4-29578)

Dcm now reports security events to IdsM according to AUTOSAR 20-11.

Support of SAE J1979-2 in Dcm and Dem (MSR4-29643)

The Dem add-on OBD on UDS and the Dcm add-on OBD now support OBDonUDS as specified in SAE J1979-2.

Additional Information
Refer to the following video for more information.

DEXT Export/Import for OBDonUDS/WWH-OBD (MSR4-29727)

The DEXT export and import now also contains content for WWH-OBD or OBDonUDS. The CDD-Import now also contains content for OBDonUDS.

DEXT Export DiagnosticSession Settings (MSR4-30168)

The DEXT export now contains all the following DiagnosticSession settings:

- DiagnosticSession.jumpToBootLoader
- DiagnosticSession.p2ServerMax
- DiagnosticSession.p2StarServerMax

The CANdelaStudio Help provides information on how to set this parameters.

Postbuild-Selectable for OBD $09 VehicleInfoTypes (MSR4-30214)

Diagnostics

The Dcm now supports the Postbuild-Selectable capable parameter DcmDspVitUsed which simplifies the variance configuration.

Supplemental Monitor Activity Data Support in Dem Add-On OBD on UDS (MSR4-30446)

The Dem add-on OBD on UDS now provides the supplemental monitor activity data as specified in SAE J1979-2.

OBD on UDS Freeze Frames (MSR4-30496)

The Dem add-on OBD on UDS now provides the storage of the OBD Freeze Frame 0x00 (First Occurrence) and 0xF0 (Most Recent) depending on the transition of TestFailedThisDcy bit as specified in SAE J1979-2.

Circular Buffer Behavior of Snapshot Records in Dem (MSR4-967)

The Dem now supports a circular buffer behavior of snapshot records. If circular buffer behavior is configured, the oldest snapshot records are overwritten by the newest snapshot records when the buffer is full.

Hardware Security Module

Secure Boot Support for Multi-Partition Use-Case (MSR4-31224)

The vHsm now also supports secure boot when multiple partitions are used. Software updates can be written to inactive partitions and an external segment list is used to verify the flash contents.

Custom Key Element for Accessing HSM2HOST and HOST2HSM Status Register (MSR4-31581)

The vHsm now provides the possibility to access the status information of the HSM2HOST and HOST2HSM register during runtime via the Crypto_30_vHsm_KeyElementGet and Crypto_30_vHsm_KeyElementSet API. For this purpose, the new custom key elements CRYPTO_KE_CUSTOM_VHSM_STATUS_HSM2HOST and CRYPTO_KE_CUSTOM_VHSM_STATUS_HOST2HSM were created for accessing the status registers.

Measurement and Calibration

Harmonization of Rtm Timing Metrics (MSR4-30445)

The timing metrics of the Rtm have been harmonized with the TA Tool Suite timing metrics as follows:

- Response Time changed to Gross Execution Time.
- ExecutionTime_Nested changed to Flat Execution Time.

Measurement and Calibration

> ExecutionTime_NonNested changed to Net Execution Time.

Complementary DAQ Channel in Xcp Slave (MSR4-30774)

The XCP Slave now supports a complementary DAQ channel which can be used by e.g. a PCIe transport layer

Memory

Multicore / Multipartition Support of vMem Memory Solution (MSR4-29669)

The vMem memory solution has been extended by the Master-Satellite concept to allow the synchronization of NV RAM data access between 1 to n-Partitions. Use-cases can be the synchronization between HSM and Application partition.

FIs and vMem Support in If_vSyncMem (MSR4-30768)

Both memory access layers, vMemAccM/vMem and AUTOSAR FIs, are now supported in the If_vSyncMem so that no re-integration is required in case the memory access layer is changed during the project.

Runtime Environment

Mode Disabling at NvBlockSwComponent (MSR4-29770)

The Rte now also supports mode disabling for runnables of NvBlock SWCs.

Variant Handling in Rte Extended (MSR4-30061)

The variant handling (IDM - Postbuild-Selectable) support of the Rte has been extended. Non-queued receivers can now receive data from an internal sender in one variant and different ECUs in the other variants.

RTE_PTR2ARRAYBASETYPE_PASSING Does Not Change Function Definitions (MSR4-30348)

The define RTE_PTR2ARRAYBASETYPE_PASSING was adapted so that double-definitions of functions cannot arise. These double-definitions sometimes caused problems with some compilers and static code analysis tools, even though the resulting assembly code was the same.

Configuration of NvM Application Ports Without Port-Defined Arguments Supported (MSR4-30771)

The Rte now supports to configure NvM application ports without port-defined arguments and to use an additional parameter for NvMBlockId instead. With this, the resource consumption can be decreased.

Background Trigger In BSW/RTE Tasks Supported (MSR4-30908)

Runtime Environment
The Rte now supports BSW/RTE tasks that have no wait points and only background triggered Runnables also in the VTT use case.
Rte RIP Support for Explicit S/R (Beta) (MSR4-31180) The Rte now supports the RTE Implementation Plugin (RIP) mechanism according to AUTOSAR for explicit non-queued sender-receiver communication as Beta. Up to now, the cross-cluster communication and general configuration of the RIPs via FlatMap is supported.
Rte Add-On Let Available (Beta) (MSR4-31198) The Rte add-on Let (Logical Execution Time) according to AUTOSAR 19-11 is now available as Beta. LET defines time intervals in which Runnables exchange its data within. When a Runnable is mapped to an LET interval, all its implicit sender/receiver data will be exchanged with LET semantics.
Rte RIP Support for Implicit S/R (Beta) (MSR4-31289) The Rte now supports the RTE Implementation Plugin (RIP) mechanism according to AUTOSAR for implicit sender/receiver communication as Beta. When the plugin is enabled, the Rte_IWrite , Rte_IRead, and Rte_IWriteRef APIs call the plugin functions and the RTE disables the buffer generation.
Rte RIP Support for Cross-Cluster Mode Handling (Beta) (MSR4-31290) The Rte now supports the RTE Implementation Plugin (RIP) mechanism according to AUTOSAR for cross-cluster mode handling as Beta.
Rte RIP Support for Explicit S/R, Implicit S/R, and Cross-Cluster Mode Handling (MSR4-31291) The Rte now supports the RTE Implementation Plugin (RIP) mechanism for Explicit S/R, Implicit S/R, and Cross-Cluster Mode Handling.
Rte Support for Provided Server on Several VLANs (MSR4-31934) The Rte now allows that client-server operations are called from several VLANs.
Runnable Activation Reasons for Multi-Core and Memory Protection (MSR4-933) The Rte now also supports the evaluation of runnable activation reasons over partition boundaries in multi-core systems and systems with memory protection.
Change of Parameters for the VFB Trace Hooks With Arrays (RTE-4237) The parameters for the VFB trace hooks will change to the format of the APIs as specified in AUTOSAR: Parameters for arrays are expected to be a pointer to the array base type. This affects all safety related Rte APIs with arrays.

Runtime Environment

This feature requires the migration of existing projects:

If the define RTE_PTR2ARRAYTYPE_PASSING is defined before Rte_Hook.h is included, the previous version of the prototypes with pointers to the arrays can be reenabled.

Safety

SafeBSW for vIpc (Memory-Based) (MSR4-240)

The moduls vIpc, vIpcMemIf and vIpcMemCp are now available for SafeBSW up to ASIL D.

Support of E2e Profile 8 (MSR4-30362)

E2e now supports profile 8 according to AUTOSAR 20-11 up to ASIL D.

Support of E2e Library Profile 44 in E2eXf (MSR4-30363)

The E2eXf now supports the E2e library profile 44 according to AUTOSAR 20-11 up to ASIL D.

Security

KeyM HSM Certificate Handling (MSR4-30372)

The KeyM now supports processing of incoming service requests by forwarding them to the MICROSAR HSM. The KeyM on application side then acts solely as proxy and any certificate operations are processed completely on HSM. The HSM provides the endpoint to the KeyM proxy.

Documentation in SIP
Refer to the TechnicalReference_KeyM.pdf for more information on the remote handling of certificates.

Save and Restore Context for Running Crypto Operations (MSR4-31232)

LibCv, Csm, and vHsm now support the save context and restore context mode for running crypto operations. With this mode, the streaming mode is interrupted after start or update to continue it again later.

Support of VKMS2.7 (MSR4-31360)

The VKMS now supports the specification Vehicle Key Management System (VKMS), version 2.7, Baseline 0.7b.

SA Selection of TcpIp Add-On IPsec Improved (MSR4-31561)

Security

The TcpIp Add-On IPsec now provides a deterministic SA (Security Association) selection if multiple matching SAs are currently part of the SA database. Now, always the last recently added SA is selected. This does not apply to IKE responder if no traffic has been observed on the SA or a configurable timeout from SA installation has not passed.

New Callout Before Jumping to Updater (MSR4-31760)

The vHsm now provides a new callout function declaration vHsm_FirmwareUpdate_PreJmpToUpdater_Callout. This callout is called at the last point before the vHsm jumps into the vHsmUpd to perform the update. This is required e.g. if an Os is used and the SysTick-Interrupt needs to be disabled before jumping to the updater.

ECBD Support in Crypto (Sw) (MSR4-949)

Crypto (Sw) now supports ECBD (Elliptic Curve Burmester Desmedt) with SECp224r1 and NIST P-224.

Software Download

Support PB-Selectable for Parent/Child Role (MSR4-28967)

The vSwUpdM now supports Postbuild-Selectable for selecting the role (Parent or Child) on startup.

Decompression for OTA Download (Vector) (MSR4-29267)

The vFotaH now supports a Vector decompression algorithm to reduce the transfer size of the software update data.

System

ComM Service Interface ComM_UserRequest Extended for PNC (MSR4-31281)

The ComM service interface ComM_UserRequest has been extended to also get the PNC state of the mapped ComMUser as specified in AUTOSAR R20-11.

Tooling

Support for RTE Implementation Plugin (DVCFG-5761)

First introduction of the RTE Implementation Plugins:

- Introduce Annotation DV:RIPS and Extend Model Abstraction.
- Introduce a validator to synchronize external RIPS FlatInstanceDescriptors with EcuExtract references.
- Provide a UI to create and modify RIPS references at port elements.

Tooling

Support for Logical Execution Time (LET) Modelling (DVCFG-6819)

First support for Logical Execution Time Modelling:

- Introduce new 'Timing' editor domain.
- Introduce editor 'Logical Execution Time' in new 'Timing' domain.

Migration Steps

General

MemMap Generator Available And Module-specific MemMap Includes (MSR4-265, MSR4-31200, MSR4-30113)

Custom-specific adaptions in the MemMap.h for these modules must be removed from the general MemMap.h and configured in DaVinci Configurator Pro.

Note
Refer to the Vector KnowledgeBase article for more information: https://support.vector.com/kb?id=kb_article_view&sysparm_article=KB0013394 .

Feature description: **MemMap Generator Available And Module-specific MemMap Includes (MSR4-265, MSR4-31200, MSR4-30113)**

Communication

Configuration of Multiple MainFunctions in Com (COM-2215)

With the support of the configuration of multiple MainFunctions, the names of the MainFunctions change and the MainFunction to Task Mapping is lost. If you want to preserve the MainFunction to Task Mapping, you must do the following steps for the migration of your configuration:

> Do the migration of your configuration.
> Run the script task RestoreReferencesRTEEVentToTaskMappings before you update the BswInternalBehavior of the Com.

Feature description: **Configuration of Multiple MainFunctions in Com (COM-2215)**

Support for DoIP Extension CONC_649 (MSR4-29025)

Due to introduction of a new DoIPInterface container some manual corrections have to be applied when migrating a configuration created with prior MICROSAR Releases and performing Input file Updates:

1) Open configuration in MSR4-R27 SIP and execute SIP migration. There will be a new container DoIP/DoIPConfigSet/DoIPInterface.

2) Perform Update of Input Files as suggested by DaVinci tool.

3) Merge content of the DoIPInterface containiner mentioned in 1) with the derived container.

Communication

4) Delete the DoIPInterface container mentioned in 1).

➡ Feature description: **Support for DoIP Extension CONC_649 (MSR4-29025)**

Option To Disable IPv4 UDP Checksum Calculation in TcpIp (MSR4-32103, TCPIP-4355)

With the introduction of this AUTOSAR feature, the handling of received/transmitted IPv4 UDP frames with checksum == 0 is changed: Received IPv4 UDP frames with checksum == 0 will be discarded by the TcpIp implementation and not forwarded to the upper layer.

To allow such frames, the functionality must be unlocked per socket:

> For SoAd as upper layer by disabling parameter:

/SoAd/SoAdConfig/SoAdSocketConnectionGroup/SoAdSocketProtocol/SoAdSocketUdp/SoAdSocketUdpChecksumEnabled

> For customer applications as upper layer by usage of TcpIp_ChangeParameter() API.

If the functionality is unlocked, the TcpIp module will also use checksum == 0 for transmission of IPv4 UDP packets on the corresponding socket(s).

See AUTOSAR requirements SWS_TCPIP_00279 and SWS_TCPIP_00280 and technical documentation for further details.

➡ Feature description: **Option To Disable IPv4 UDP Checksum Calculation in TcpIp (MSR4-32103, TCPIP-4355)**

Runtime Environment

Change of Parameters for the VFB Trace Hooks With Arrays (RTE-4237)

If the define RTE_PTR2ARRAYTYPE_PASSING is defined before Rte_Hook.h is included, the previous version of the prototypes with pointers to the arrays can be reenabled.

➡ Feature description: **Change of Parameters for the VFB Trace Hooks With Arrays (RTE-4237)**